Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) -- YEAR 10 SEMESTER 2

ASSESSMENT MATERIALS & REPORTING PACK

Academic Year: 2025-2026 (Second Semester)

Course Code: CIE IGCSE Computer Science (0478)

This Pack Contains: 4 Full Mock Examinations + Mark Schemes + Grade Boundaries

+ Gap Analysis Templates + Target Sheets + Report Templates

+ Exam Day Guide + Bilingual Glossary

MOCK EXAM 2 -- PAPER 1 (WEEK 9)

THEORY OF COMPUTER SCIENCE

Time: 1 hour 45 minutes

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

- Write your name, class, and date in the spaces provided.
- Answer ALL questions.
- Write your answers in the spaces provided.
- If you need extra space, ask for additional paper.
- Calculators are NOT permitted.

ADVICE TO CANDIDATES

- Read each question carefully before answering.
- Pay attention to the command words (State, Describe, Explain, Evaluate).
- Show your working for all calculation questions.
- Check your answers if you have time at the end.

SECTION A: SHORT ANSWER QUESTIONS (30 marks)

Answer ALL questions in this section.

QUESTION 1 (Data Representation)

(a) Convert the denary number 91 to 8-bit binary. [1]

(b) Convert the binary number 11010110 to hexadecimal. [2]

(c) Explain why a programmer might prefer to use hexadecimal rather than binary when debugging memory contents. [2]

(d) A sound file is recorded with a sample rate of 44.1 kHz and a sample resolution of 16 bits. The recording lasts for 3 minutes.

Calculate the file size of the uncompressed sound file in megabytes (MB).

Show your working. [3]

[Question 1 Total: 8 marks]

QUESTION 2 (Hardware)

(a) Name the component in the CPU that performs arithmetic and logical operations. [1]

(b) State the purpose of each of the following registers:

(i) Program Counter (PC) [1]

(ii) Memory Data Register (MDR) [1]

(c) Describe the fetch-decode-execute cycle. In your answer, explain the role of the buses (address bus, data bus, control bus). [4]

(d) Explain why having more cache memory generally improves the performance of a computer. [2]

[Question 2 Total: 9 marks]

QUESTION 3 (Software)

(a) State two functions of an operating system. [2]

1. ____
2. ____

(b) Describe the difference between a compiler and an interpreter. [3]

(c) Give two differences between freeware and shareware. [2]

3. ____
4. ____

(d) Explain one advantage of using open source software for a school. [2]

[Question 3 Total: 9 marks]

QUESTION 4 (The Internet and Security)

(a) Draw and label a diagram of a star network topology with 4 computers and one central connecting device. [2]

[Draw your diagram in this space]

(b) Describe the role of a router in a network. [2]

(c) Explain how a DNS server is used when a user types a URL into a web browser. [3]

(d) State two security threats to computer systems and, for each, describe one method of protection. [4]

Threat 1: ____

Protection: ____

Threat 2: ____

Protection: ____

[Question 4 Total: 11 marks]

SECTION A TOTAL: 37 marks

SECTION B: STRUCTURED QUESTIONS (30 marks)

Answer ALL questions in this section.

QUESTION 5 (Data Representation and Compression)

An image has a resolution of 1024 x 768 pixels and uses 24-bit colour depth.

(a) Calculate the file size of the uncompressed image in kilobytes (KB).

Show your working. [3]

(b) The image is compressed using lossless compression, reducing the file size by 40%.

Calculate the compressed file size in KB. [2]

(c) Explain the difference between lossy and lossless compression.

Give one example of when each would be appropriate. [4]

(d) A 16-bit sample resolution is used instead of 8-bit for recording sound.

Explain the effect this has on:

(i) The quality of the recorded sound [2]

(ii) The file size of the recording [1]

(i) ____

(ii) ____

[Question 5 Total: 12 marks]

QUESTION 6 (Networks, Internet Technologies and Security)

A company has offices in Beijing and Shenzhen. Employees need to share files and access a central database. The IT manager is designing the network.

(a) The manager decides to use a Wide Area Network (WAN) to connect the offices.

Explain why a WAN is appropriate for this situation. [2]

(b) Describe the difference between a LAN and a WAN. [2]

(c) The manager wants to ensure secure communication between the offices.

Describe two methods that could be used to keep the data secure during transmission. [4]

Method 1: ____

Method 2: ____

(d) The company website collects customer information including names, addresses and payment details.

(i) Explain the purpose of a cookie on a website. [2]

(ii) Describe one privacy concern related to the use of cookies. [2]

(e) The IT manager is considering using cloud storage for company files.

Evaluate the use of cloud storage for this company. [6]

[Question 6 Total: 18 marks]

SECTION B TOTAL: 30 marks

SECTION C: EXTENDED ANSWER QUESTION (15 marks)

Answer this question.

QUESTION 7 (Emerging Technologies and Automated Systems)

Autonomous vehicles (self-driving cars) are being developed by several companies. These vehicles use a combination of sensors, artificial intelligence, and GPS to navigate without human input.

(a) Sensors in autonomous vehicles collect data about the surrounding environment.

(i) Name two types of sensors that would be used in an autonomous vehicle and state what each would detect. [4]

Sensor 1: ____

Detects: ____

Sensor 2: ____

Detects: ____

(ii) Explain the role of an Actuator in an autonomous vehicle. [2]

(b) Describe what is meant by artificial intelligence (AI). [2]

(c) Expert systems can be used to help diagnose problems in vehicles. Describe the three main components of an expert system. [3]

(d) Evaluate the potential impact of autonomous vehicles on society. Your answer should consider both positive and negative effects. [4]

[Question 7 Total: 15 marks]

SECTION C TOTAL: 15 marks

PAPER TOTAL: 82 marks (raw, before standardisation to 75)

Note to teacher: Standardise to 75 marks by scaling: (student mark / 82) x 75

END OF MOCK EXAM 2 -- PAPER 1

MOCK EXAM 2 -- PAPER 1: MARK SCHEME

SECTION A: SHORT ANSWER QUESTIONS

QUESTION 1 (Data Representation) -- 8 marks

(a) 91 in 8-bit binary [1]

ACCEPT: 01011011

REJECT: 1011011 (7-bit, no mark)

(b) 11010110 to hexadecimal [2]

Working: 1101 = D, 0110 = 6 [1]

Answer: D6 [1]

(Allow follow-through for correct conversion of their 4-bit groupings)

(c) Why use hexadecimal for debugging [2]

- Hexadecimal uses fewer digits than binary [1]
- Easier for humans to read and remember / less prone to errors [1]
- Hex is a shorthand for binary (each hex digit = 4 bits) [1]

(Any 2 valid points for 2 marks)

ACCEPT equivalent wording

(d) Sound file size calculation [3]

Sample rate = 44.1 kHz = 44100 Hz

Duration = 3 minutes = 180 seconds

Resolution = 16 bits

File size = 44100 x 16 x 180 [1 for method]

= 127,008,000 bits

= 15,876,000 bytes [1]

= 15.14 MB (accept 15.1 MB or 15 MB) [1]

ALTERNATIVE working (using kHz directly):

File size = 44.1 x 1000 x 16 x 180 = 127,008,000 bits

(Accept answer in bytes: 15,876,000 B or approximately 15.1 MB)

(Allow follow-through for arithmetic errors)

QUESTION 2 (Hardware) -- 9 marks

(a) Component performing arithmetic/logical operations [1]

ACCEPT: ALU / Arithmetic Logic Unit

REJECT: CPU (too vague)

(b)(i) Purpose of Program Counter [1]

ACCEPT: Holds the address of the next instruction to be fetched

ACCEPT: Keeps track of where the program has reached

REJECT: Stores instructions (confused with CIR)

(b)(ii) Purpose of Memory Data Register [1]

ACCEPT: Holds data being transferred to/from memory

ACCEPT: Temporarily stores data fetched from or to be written to memory

ACCEPT: Holds the contents of the memory address being accessed

(c) Fetch-decode-execute cycle [4]

Award marks for any 4 of:

- PC contains address of next instruction [1]
- Address copied to MAR along address bus [1]
- Instruction copied from memory to MDR along data bus [1]
- Instruction copied from MDR to CIR [1]
- Instruction decoded by CU [1]

- Instruction executed (by ALU or other components) [1]
- PC incremented to point to next instruction [1]

Award 1 mark for mentioning the three buses:

- Address bus carries memory addresses (one-way) [1]
- Data bus carries data/instructions (two-way) [1]
- Control bus sends control signals (read/write) [1]

(Max 4 marks total)

(d) Why more cache improves performance [2]

- Cache is faster than RAM / closer to CPU speed [1]
- CPU checks cache before RAM [1]
- If data is in cache (cache hit), it is accessed much faster [1]
- Reduces bottleneck between CPU and RAM [1]

(Any 2 valid points for 2 marks)

QUESTION 3 (Software) -- 9 marks

(a) Two functions of an OS [2]

Award 1 mark each for any 2 of:

- Memory management
- File management
- Process management / multitasking
- Security management
- User interface provision
- Device management / peripheral control
- Error reporting

(Must be different functions for 2 marks)

(b) Compiler vs interpreter [3]

Award marks for:

- Compiler translates entire source code before execution [1]
- Interpreter translates and executes line by line [1]
- Compiler produces a standalone executable file [1]
- Interpreter does not produce an executable / needs re-interpreting each time [1]
- Compiler reports all errors at the end [1]
- Interpreter reports errors immediately when found [1]
- Compiler produces faster execution [1]
- Interpreter allows easier debugging [1]

(Any 3 distinct, valid points for 3 marks)

ACCEPT equivalent wording

(c) Two differences between freeware and shareware [2]

Award 1 mark each for any 2 valid differences:

- Freeware is completely free; shareware has a trial period [1]
- Freeware has no payment; shareware requires payment for full version [1]
- Freeware may have limited functionality; shareware has all features in trial [1]

- Freeware is fully available; shareware may expire [1]

(Must be differences, not just two facts about each)

(d) Advantage of open source for a school [2]

Award marks for:

- Free to use / no licensing costs [1]
- Source code can be modified to suit school's needs [1]
- Large community for support and updates [1]
- Students can learn from reading the source code [1]
- Not dependent on a single vendor [1]

(1 mark for identifying advantage, 1 mark for explanation/elaboration)

QUESTION 4 (Internet and Security) -- 11 marks

(a) Star network diagram [2]

Award marks for:

- Central device drawn (switch/hub/server) [1]
- Four computers/devices connected to central device [1]
- Correct topology (star shape with central node) [1]

(Max 2 marks -- 1 for central device, 1 for 4 computers connected correctly)

ACCEPT hand-drawn or described in text if diagram poor

(b) Role of a router [2]

- Connects different networks together [1]
- Routes data packets to their destination using IP addresses [1]
- Directs traffic between networks [1]

(Any 2 valid points for 2 marks)

(c) How DNS works [3]

- User types URL (human-readable domain name) [1]
- Browser sends request to DNS server [1]
- DNS server looks up the domain name and returns the corresponding IP address [1]
- Browser uses IP address to connect to the web server [1]

(Any 3 valid, sequential points for 3 marks)

(d) Two threats and protections [4]

Award 1 mark for each threat + 1 mark for each matching protection (max 4)

THREATS (any 2):

- Virus [1] Protection: Antivirus software [1]
- Worm [1] Protection: Firewall / Antivirus [1]
- Trojan [1] Protection: Antivirus / user education [1]
- Phishing [1] Protection: User education / email filters [1]
- Hacking [1] Protection: Firewall / strong passwords / encryption [1]
- Spyware [1] Protection: Antivirus / anti-spyware [1]
- Pharming [1] Protection: Check URL / use trusted DNS [1]

- Malware [1]   Protection: Antivirus / firewall [1]

ACCEPT any valid threat-protection pair

Protection must match threat for mark

SECTION B: STRUCTURED QUESTIONS

QUESTION 5 (Data Representation and Compression) -- 12 marks

(a) File size calculation [3]

Resolution = 1024 x 768 = 786,432 pixels [1]

File size = 786,432 x 24 bits [1]

= 18,874,368 bits

= 2,359,296 bytes

= 2,304 KB (or approximately 2304 KB) [1]

ALTERNATIVE:

1024 x 768 x 24 / 8 = 2,359,296 bytes [1 for dividing by 8]

2,359,296 / 1024 = 2,304 KB [1]

(Allow follow-through for arithmetic errors)

(b) Compressed file size [2]

40% reduction means file is 60% of original [1]

2304 x 0.60 = 1,382.4 KB (accept 1382 KB) [1]

OR: 2304 - (2304 x 0.40) = 1382.4 KB

(Allow follow-through from part (a))

(c) Lossy vs lossless [4]

Lossy compression:

- Permanently removes some data from the file [1]
- File cannot be restored to exact original [1]
- Example: JPEG images, MP3 audio [1]

Lossless compression:

- Reduces file size without losing any data [1]
- Original file can be perfectly restored [1]
- Example: PNG images, ZIP files, FLAC audio [1]

Example appropriateness:

- Lossy appropriate for: photographs where slight quality loss is acceptable [1]
- Lossless appropriate for: text documents, medical images where every detail matters [1]

(Max 4 marks -- must cover both types with one example each)

(d)(i) Effect of 16-bit vs 8-bit on quality [2]

- Greater sample resolution means more bits to represent each sample [1]
- More accurate representation of the sound wave [1]
- Higher fidelity / better quality / less quantisation error [1]

- Greater dynamic range [1]

(Any 2 valid points)

(d)(ii) Effect on file size [1]

- File size will be doubled / larger [1]
- 16-bit uses twice as much storage per sample compared to 8-bit [1]

(Any 1 valid point)

QUESTION 6 (Networks, Internet Technologies and Security) -- 18 marks

(a) Why WAN is appropriate [2]

- Offices are in different cities (geographically separated) [1]
- WAN connects devices over large distances / wide area [1]
- Allows sharing of resources between remote locations [1]

(Any 2 valid points)

(b) LAN vs WAN [2]

Award 1 mark for each valid difference (max 2):

- LAN covers small area (building/campus) vs WAN covers large area (cities/countries) [1]
- LAN is usually owned by one organisation vs WAN uses third-party infrastructure [1]
- LAN has higher data transfer rates vs WAN is generally slower [1]
- LAN has lower setup cost vs WAN is more expensive [1]
- LAN has fewer errors vs WAN has more transmission errors [1]

(c) Two security methods for data transmission [4]

Method 1 -- Encryption [1]

- Data is scrambled so it cannot be read if intercepted [1]
- Uses keys to encrypt at sender and decrypt at receiver [1]

Method 2 -- SSL/TLS [1]

- Creates a secure encrypted connection between computers [1]
- HTTPS uses SSL/TLS to protect data in transit [1]

ALTERNATIVE Method 2 -- VPN [1]

- Creates a private tunnel through public internet [1]
- Encrypts all data sent between offices [1]

ALTERNATIVE Method 2 -- Firewall [1]

- Filters incoming/outgoing network traffic [1]
- Blocks unauthorised access between networks [1]

(Any 2 valid methods with explanation, max 4 marks)

(d)(i) Purpose of a cookie [2]

- Small text file stored on user's computer by a website [1]
- Used to remember information about the user / track browsing behaviour [1]
- Stores preferences / login details / shopping cart contents [1]
- Allows website to recognise returning visitors [1]

(Any 2 valid points)

(d)(ii) Privacy concern with cookies [2]

- Cookies can track user's browsing habits across multiple websites [1]
- Personal information may be collected without explicit consent [1]
- Data may be sold to third parties for targeted advertising [1]
- Users may not be aware of what data is being stored [1]

(1 mark for concern, 1 mark for elaboration)

(e) Evaluation of cloud storage [6]

Award marks using levels approach:

Level 3 (5-6 marks): Detailed evaluation with balanced view

- Clear explanation of cloud storage (data stored on remote servers accessed via internet) [1]
- At least TWO advantages explained [2]

* Accessible from any location with internet

* No need for local storage infrastructure

* Automatic backups

* Easy file sharing between offices

* Scalable (can increase storage as needed)

* Cost-effective (no hardware maintenance)

- At least TWO disadvantages explained [2]

* Requires internet connection

* Security concerns (data held by third party)

* Ongoing subscription costs

* Data privacy and compliance issues

* Risk of data breach

- Clear conclusion or balanced judgement [1]

Level 2 (3-4 marks): Some evaluation with limited balance

- Basic explanation of cloud storage [1]
- One or two advantages mentioned [1-2]
- One or two disadvantages mentioned [1-2]
- Limited or no conclusion

Level 1 (1-2 marks): Basic description only

- Simple description of cloud storage [1]
- One advantage OR one disadvantage [1]
- No real evaluation

Level 0 (0 marks): No relevant content

SECTION C: EXTENDED ANSWER QUESTION

QUESTION 7 (Emerging Technologies and Automated Systems) -- 15 marks (a)(i) Two sensors in autonomous vehicle [4]

Award 1 mark for sensor name + 1 mark for what it detects (x2 pairs):

- Light sensor / camera [1] -- detects road markings, traffic lights, other vehicles, obstacles [1]
- Proximity / ultrasonic sensor [1] -- detects distance to objects, parking assistance [1]
- Temperature sensor [1] -- detects engine temperature, road conditions [1]
- Speed sensor [1] -- detects vehicle speed, wheel rotation [1]
- GPS receiver [1] -- detects vehicle location, navigation [1]
- Accelerometer [1] -- detects motion, sudden braking, collisions [1]
- RADAR / LIDAR [1] -- detects objects at distance, mapping environment [1]
- Gyroscope [1] -- detects orientation, balance [1]

(Any 2 valid sensor + detection pairs, max 4 marks)

(a)(ii) Role of actuator [2]

- An actuator converts electrical signals into physical movement [1]
- In an autonomous vehicle, actuators control steering, braking, acceleration [1]
- Actuators carry out the decisions made by the computer [1]

(Any 2 valid points)

(b) Artificial intelligence [2]

- Computer systems that can perform tasks that normally require human intelligence [1]
- Systems that can learn from experience / adapt to new data [1]
- Ability to recognise patterns, make decisions, solve problems [1]
- Examples: speech recognition, image recognition, decision making [1]

(Any 2 valid points)

(c) Expert system components [3]

Award 1 mark for each:

- Knowledge base: stores facts and information about the domain [1]
- Inference engine: applies rules to the knowledge base to reach conclusions [1]
- Rules base: contains IF-THEN rules used to make decisions [1]
- User interface: allows user to interact with the system [1]

(Any 3 valid components for 3 marks)

(d) Evaluate impact of autonomous vehicles [4]

Award marks for balanced evaluation:

Positive impacts (up to 2 marks):

- Reduced accidents due to elimination of human error [1]
- Improved mobility for elderly and disabled people [1]
- Reduced traffic congestion through optimised routes [1]

- Increased productivity (passengers can work while travelling) [1]
- Reduced need for parking spaces in cities [1]
- Lower emissions through efficient driving [1]

Negative impacts (up to 2 marks):

- Job losses for professional drivers [1]
- Ethical dilemmas in accident situations [1]
- Security risks (hacking of vehicle systems) [1]
- High development and purchase costs [1]
- Legal and insurance complications [1]
- Dependence on technology / loss of driving skills [1]
- Privacy concerns (tracking of journeys) [1]

(Max 4 marks -- at least 1 positive and 1 negative for full marks)

GRADE BOUNDARIES -- MOCK EXAM 2 PAPER 1

RAW MARKS (out of 82, scale to 75):

Grade	Minimum Raw Mark (/82)	Scaled Mark (/75)
A*	66	60
A	57	52
B	49	45
C	41	37
D	33	30
E	25	23
F	17	15
U	0-16	0-14

SCALING FORMULA: Scaled Mark = round((Raw Mark / 82) x 75)

Alternative (if not scaling): Use raw marks out of 82 with these boundaries:

Grade	Minimum Mark (/82)
A*	66
A	57
B	49
C	41
E	25
F	17
U	0-16

MOCK EXAM 3 -- PAPER 2 (WEEK 10)

PROBLEM-SOLVING AND PROGRAMMING

Time: 1 hour 45 minutes

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

- Write your name, class, and date in the spaces provided.
- Answer ALL questions.
- Write your answers in the spaces provided.
- If you need extra space, ask for additional paper.
- Calculators are NOT permitted.

ADVICE TO CANDIDATES

- Read each question carefully before answering.
- For pseudocode questions, use standard pseudocode syntax.
- For trace table questions, show all working.
- Read scenario information carefully before attempting Section C.

PSEUDOCODE REFERENCE (provided in exam)

```
DECLARE <variable> : <type>
CONSTANT <name> = <value>
<identifier> <-- <value>
INPUT <identifier>
OUTPUT <identifier/value/string>
IF <condition> THEN … ELSE … ENDIF
CASE OF <identifier>:
  <value>: <statement(s)>
  OTHERWISE: <statement(s)>
ENDCASE
FOR <identifier> <-- <value1> TO <value2> STEP <value> … NEXT
REPEAT … UNTIL <condition>
WHILE <condition> DO … ENDWHILE
FUNCTION <name>(<parameters>) RETURNS <type> … ENDFUNCTION
PROCEDURE <name>(<parameters>) … ENDPROCEDURE
RETURN <value>
CALL <name>(<parameters>)
COMMENT // <text>
AND, OR, NOT
LENGTH(<string>)
SUBSTRING(<string>, <start>, <length>)
ASC(<character>)
RANDOM()
ROUND(<number>, <decimal places>)
MOD, DIV
>=, <=, <>, =, <, >
```

SECTION A: SHORT ANSWER QUESTIONS (25 marks)

Answer ALL questions in this section.

QUESTION 1 (Algorithms and Program Design)

(a) State two characteristics of a good algorithm. [2]

5. ____
6. ____

(b) Describe what is meant by the term "decomposition" in algorithm design. [2]

(c) An algorithm needs to find the largest number from a list of 100 numbers.

Describe the steps this algorithm would follow. You do not need to write pseudocode. [3]

[Question 1 Total: 7 marks]

QUESTION 2 (Programming Constructs)

(a) Look at this pseudocode:

```
FOR Counter <-- 1 TO 10
  OUTPUT Counter * 3
NEXT
```

(i) State the output of this algorithm. [1]

(ii) State the data type of the variable Counter. [1]

(b) Look at this pseudocode:

```
DECLARE Password : STRING
DECLARE Valid : BOOLEAN
Valid <-- FALSE
REPEAT
  OUTPUT "Enter password (at least 8 characters):"
  INPUT Password
  IF LENGTH(Password) >= 8 THEN
    Valid <-- TRUE
  ELSE
    OUTPUT "Password too short"
  ENDIF
UNTIL Valid = TRUE
OUTPUT "Password accepted"
```

(i) State the purpose of this algorithm. [1]

(ii) State the loop structure used and explain why it is appropriate for this task. [2]

(c) Look at this pseudocode:

```
DECLARE x, y : INTEGER
x <-- 10
y <-- 3
OUTPUT x DIV y
OUTPUT x MOD y
```

State the two outputs of this algorithm. [2]

Output 1: ____

Output 2: ____

(d) Explain the difference between a function and a procedure. [2]

[Question 2 Total: 9 marks]

QUESTION 3 (Databases and SQL)

A school uses a relational database to store information about students and their exam results. The database has the following tables:

STUDENT (StudentID, FirstName, LastName, Class, DateOfBirth)

RESULT (ResultID, StudentID, Subject, Mark, Grade)

(a) Explain why this is a relational database rather than a flat-file database. [2]

(b) State the primary key of the STUDENT table. [1]

(c) Explain the purpose of the StudentID field in the RESULT table. [2]

(d) Write an SQL statement to display the first name and last name of all students in Class "10A". [2]

(e) Write an SQL statement to count how many students are in each class. The output should show the class name and the number of students. [3]

[Question 3 Total: 10 marks]

SECTION A TOTAL: 26 marks

SECTION B: CODE READING AND WRITING (35 marks)

Answer ALL questions in this section.

QUESTION 4 (Trace Tables and Code Reading)

Study the following pseudocode:

```
DECLARE Total, Count, Number : INTEGER
Total <-- 0
Count <-- 0
REPEAT
  OUTPUT "Enter a positive number (0 to stop):"
  INPUT Number
  IF Number > 0 THEN
    Total <-- Total + Number
```

```
    Count <-- Count + 1
   ENDIF
  UNTIL Number = 0
  IF Count > 0 THEN
   OUTPUT "Average is", Total / Count
  ELSE
   OUTPUT "No numbers entered"
  ENDIF
```

(a) Complete the trace table for the following input sequence: 5, 8, 3, 0

The first row has been completed for you. [6]

Number	Total	Count	Number > 0	Number = 0	OUTPUT
--	0	0	--	--	prompt text

(b) State the final output of the algorithm with the input sequence 5, 8, 3, 0. [1]

(c) State what would happen if the user enters 0 as the first input.

Explain why. [2]

[Question 4 Total: 9 marks]

QUESTION 5 (Pseudocode Writing -- Functions and Arrays)

(a) Write a function called CalculateAverage that takes an array of integers and the number of items in the array as parameters. The function should return the average of the numbers as a real number.

You should declare all variables used. [5]

(b) Write a procedure called FindMax that takes an array of integers called Scores and an integer called Size as parameters. The procedure should output the largest value in the array and the position (index) where it was found.

You should declare all variables used. [5]

(c) Look at this pseudocode for a linear search:

```
  DECLARE Numbers : ARRAY[1:10] OF INTEGER
  DECLARE Target, Index : INTEGER
  DECLARE Found : BOOLEAN
  Found <-- FALSE
  Index <-- 1
  WHILE Index <= 10 AND Found = FALSE DO
   IF Numbers[Index] = Target THEN
    Found <-- TRUE
   ELSE
```

```
    Index <-- Index + 1
  ENDIF
ENDWHILE
IF Found = TRUE THEN
  OUTPUT "Found at position", Index
ELSE
  OUTPUT "Not found"
ENDIF
```

(i) State the initial value of Found and explain its purpose. [2]

(ii) Explain what would happen if the array contained the Target value more than once. [2]

[Question 5 Total: 14 marks]

QUESTION 6 (Boolean Logic)

(a) Complete the truth table for the logic expression: X = A AND (NOT B)

A	B	NOT B	A AND (NOT B)
0	1		
1	0		

[2]

(b) A security system uses two inputs:

- A: Motion sensor (1 = motion detected, 0 = no motion)
- B: Keypad entry correct (1 = correct code, 0 = incorrect)

The alarm sounds (output X = 1) if motion is detected AND the correct code has NOT been entered.

(i) Write the Boolean expression for this system. [1]

X = ____

(ii) Draw the logic circuit for this system. Label all inputs and the output. [3]

[Draw in this space]

(c) Simplify the following Boolean expression. Show your working. [2]

(A AND B) OR (A AND B)

(d) A truth table has three inputs (A, B, C) and one output (X).

The output X is 1 when:

- A is 1 AND B is 1, OR
- C is 1

Write the Boolean expression for X. [1]

X = ____

[Question 6 Total: 9 marks]

SECTION B TOTAL: 32 marks

SECTION C: SCENARIO QUESTION (15 marks)

Answer this question in the spaces provided.

QUESTION 7 (Scenario)

A small library needs a computer program to manage book loans. The program must allow a librarian to:

- Enter a member's ID number (must be exactly 6 digits)
- Enter a book's ISBN (must be exactly 10 characters)
- Record the loan date
- Check if the member has any overdue books (loans more than 14 days old)
- Calculate and display the due date (14 days from loan date)
- Store the loan details in a file called "Loans.txt"
- Display a confirmation message

The librarian needs to process loans for up to 50 members per day. Each member can borrow up to 3 books at a time.

The program should validate all inputs and display appropriate error messages.

(a) (i) Design an algorithm to process one book loan. Your algorithm should:

- Validate the member ID (6 digits)
- Validate the ISBN (10 characters)
- Calculate the due date (14 days from loan date)
- Store the loan details (MemberID, ISBN, LoanDate, DueDate) in a file called "Loans.txt"
- Display a confirmation message

You should use pseudocode. You may assume the current date is available from a function called GetCurrentDate() which returns a string. [8]

(ii) Explain how your algorithm validates the member ID. [2]

(b) The library wants to check if a member has any overdue books.

Describe how a program could read the "Loans.txt" file and determine if a member has overdue books. You do not need to write pseudocode. [3]

(c) The librarian wants to limit members to borrowing a maximum of 3 books.

Explain how this restriction could be implemented in the program. [2]

[Question 7 Total: 15 marks]

SECTION C TOTAL: 15 marks

PAPER TOTAL: 73 marks (raw, before standardisation to 75)

Note to teacher: Standardise to 75 marks by scaling: (student mark / 73) x 75

END OF MOCK EXAM 3 -- PAPER 2

MOCK EXAM 3 -- PAPER 2: MARK SCHEME

SECTION A: SHORT ANSWER QUESTIONS

QUESTION 1 (Algorithms and Program Design) -- 7 marks

(a) Two characteristics of a good algorithm [2]

Award 1 mark each for any 2 of:

- Clear and unambiguous steps
- Finitely many steps (terminates)
- Each step is precisely defined
- Produces the correct output for given input
- Efficient (not unnecessary steps)
- Well-structured (uses appropriate control structures)

(b) Decomposition [2]

- Breaking a complex problem down into smaller, more manageable sub-problems [1]
- Each sub-problem can be solved independently [1]
- Solutions to sub-problems are combined to solve the overall problem [1]

(Any 2 valid points)

(c) Steps to find largest number [3]

Award marks for describing the algorithm:

- Set a variable (e.g., Largest) to the first number / a very small number [1]
- Compare each number in the list with the current Largest [1]
- If a number is larger than Largest, update Largest to that number [1]
- Continue until all numbers have been checked [1]
- Output the value of Largest [1]

(Any 3 valid, sequential steps)

(If pseudocode given instead of description, accept if correct)

QUESTION 2 (Programming Constructs) -- 9 marks

(a)(i) Output of the FOR loop [1]

ACCEPT: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30

ACCEPT: List written vertically

ACCEPT: Any clear format showing all 10 numbers

(a)(ii) Data type of Counter [1]

ACCEPT: INTEGER

REJECT: INT (not standard pseudocode)

(b)(i) Purpose of the algorithm [1]

ACCEPT: To get a password of at least 8 characters from the user

ACCEPT: To validate that a password has 8 or more characters

ACCEPT: Input validation for password length

REJECT: "Checks password" (too vague)

(b)(ii) Loop structure and why appropriate [2]

- REPEAT...UNTIL loop [1]
- Because the password must be entered at least once before checking [1]
- OR: The loop runs until valid input is received [1]
- OR: Post-condition loop means input happens before validation [1]

(1 mark for identifying REPEAT...UNTIL, 1 mark for valid reason)

(c) DIV and MOD outputs [2]

Output 1 (DIV): 3 [1]

Output 2 (MOD): 1 [1]

(Both correct for 2 marks, 1 mark if one correct)

(d) Function vs Procedure [2]

Award marks for:

- A function returns a value to the calling program [1]
- A procedure does not return a value / performs a task [1]
- A function is used in expressions / assignments [1]
- A procedure is called as a standalone statement [1]

(Any 2 distinct, valid differences)

ACCEPT equivalent wording

QUESTION 3 (Databases and SQL) -- 10 marks

(a) Why relational not flat-file [2]

- Data is stored in separate related tables [1]
- Reduces data redundancy (avoids repeating student details for each result) [1]
- Data integrity is improved [1]
- Easier to maintain and update [1]

(Any 2 valid points)

(b) Primary key of STUDENT table [1]

ACCEPT: StudentID

REJECT: "ID" (too vague, must match field name)

(c) Purpose of StudentID in RESULT table [2]

- It is a foreign key [1]
- It links each result record to the corresponding student in the STUDENT table [1]
- It establishes the relationship between the two tables [1]
- Allows data from both tables to be joined [1]

(Foreign key identification + explanation of linking for 2 marks)

(d) SQL to display first and last name of Class 10A students [2]

SELECT FirstName, LastName [1]

FROM STUDENT [1]

WHERE Class = '10A' [1]

(Max 2 marks -- 1 for correct fields, 1 for correct table and condition)

ACCEPT case-insensitive

ACCEPT double quotes around "10A"

REJECT: Missing quotes around 10A (0 marks for condition)

(e) SQL to count students per class [3]

SELECT Class, COUNT(*) [1]

FROM STUDENT [1]

GROUP BY Class [1]

ACCEPT: COUNT(StudentID)

ACCEPT: Order: SELECT, FROM, GROUP BY in any order that works

Deduct 1 mark if GROUP BY missing (max 2)

Deduct 1 mark if Class not in SELECT (max 2)

SECTION B: CODE READING AND WRITING

QUESTION 4 (Trace Tables and Code Reading) -- 9 marks

(a) Trace table [6]

Award marks row by row (allow follow-through):

Number	Total	Count	Number > 0	Number = 0	OUTPUT
--	0	0	--	--	Enter number...
5	5	1	TRUE	FALSE	prompt
8	13	2	TRUE	FALSE	prompt
3	16	3	TRUE	FALSE	prompt
0	16	3	FALSE	TRUE	prompt (hidden)
--	16	3	--	--	Average is 5.33

Row 1 (5): Total=5, Count=1, Number>0=TRUE [1]

Row 2 (8): Total=13, Count=2, Number>0=TRUE [1]

Row 3 (3): Total=16, Count=3, Number>0=TRUE [1]

Row 4 (0): Total=16, Count=3, Number>0=FALSE, Number=0=TRUE [1]

Correct final row (showing Average output) [1]

Correct intermediate column values throughout [1]

(Allow follow-through if Total/Count values cascade correctly)

(Accept "Average is 5" or "Average is 5.3" or exact division)

(b) Final output [1]

ACCEPT: "Average is 5.333..." or equivalent

ACCEPT: "Average is 5.3" (rounded)

ACCEPT: "Average is 5" (truncated)

(Must match their trace table -- allow follow-through)

(c) What happens with 0 as first input [2]

- Output would be "No numbers entered" [1]
- Because Count remains 0 (no positive numbers were entered) [1]
- The IF condition (Count > 0) is false, so ELSE executes [1]

(Correct output + correct explanation for 2 marks)

QUESTION 5 (Pseudocode Writing -- Functions and Arrays) -- 14 marks

(a) Function CalculateAverage [5]

Award marks for:

```
FUNCTION CalculateAverage(Numbers: ARRAY, Size: INTEGER) RETURNS REAL [1]
 DECLARE Sum, Index : INTEGER                    [1]
 DECLARE Average : REAL                          [1]
 Sum <-- 0                                       [1]
 FOR Index <-- 1 TO Size                         [1]
  Sum <-- Sum + Numbers[Index]                   [1]
 NEXT                                            [1]
 Average <-- Sum / Size                          [1]
 RETURN Average                                  [1]
ENDFUNCTION                                      [1]
```

(Max 5 marks -- award best 5 points)

Key marking points:

- Correct function header with return type REAL [1]
- Correct parameters (array and size) [1]
- Initialising Sum to 0 [1]
- Loop through all array elements [1]
- Adding each element to Sum [1]
- Calculating and returning average [1]
- Correct function closing [1]

ALTERNATIVE using WHILE loop: accept if correct

Deduct 1 mark if Sum not initialised

Deduct 1 mark if no RETURN statement

(b) Procedure FindMax [5]

Award marks for:

```
PROCEDURE FindMax(Scores: ARRAY, Size: INTEGER)       [1]
 DECLARE MaxValue, MaxIndex, Index : INTEGER          [1]
 MaxValue <-- Scores[1]                               [1]
 MaxIndex <-- 1                                       [1]
 FOR Index <-- 2 TO Size                              [1]
  IF Scores[Index] > MaxValue THEN                    [1]
```

```
      MaxValue <-- Scores[Index]              [1]
      MaxIndex <-- Index                      [1]
    ENDIF                                     [1]
  NEXT                                        [1]
  OUTPUT "Largest value:", MaxValue           [1]
  OUTPUT "Position:", MaxIndex                [1]
ENDPROCEDURE                                  [1]
```

(Max 5 marks -- award best 5 points)

Key marking points:

- Correct procedure header [1]
- Initialising MaxValue and MaxIndex [1]
- Loop through array from position 2 [1]
- Correct comparison and update [1]
- Output of both max value and position [1]

ALTERNATIVE: Using WHILE loop -- accept if correct

ACCEPT: Starting loop at 1 (redundant comparison but correct)

Deduct 1 mark if MaxValue not initialised

Deduct 1 mark if only value OR position output (not both)

(c)(i) Initial value of Found and its purpose [2]

- Found is initialised to FALSE [1]
- Purpose: to indicate that the target has not yet been found [1]
- OR: acts as a flag to control the loop [1]
- OR: prevents unnecessary searching once target is found [1]

(1 mark for value, 1 mark for explanation)

(c)(ii) Target appears more than once [2]

- The algorithm will find and report the FIRST occurrence [1]
- Because once Found is set to TRUE, the loop stops [1]
- It will not continue searching for additional occurrences [1]

(Correct behaviour + correct reason for 2 marks)

QUESTION 6 (Boolean Logic) -- 9 marks

(a) Truth table for X = A AND (NOT B) [2]

A	B	NOT B	A AND (NOT B)
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

Award 1 mark for correct NOT B column

Award 1 mark for correct output column

(Both must be fully correct for each mark)

(b)(i) Boolean expression for security system [1]

X = A AND (NOT B) [1]

ACCEPT equivalent: X = NOT B AND A

REJECT: Missing NOT (wrong logic)

(b)(ii) Logic circuit diagram [3]

Award marks for:

- Input A connected to AND gate [1]
- Input B connected through NOT gate first [1]
- NOT gate output connected to AND gate [1]
- AND gate output labelled X [1]

Correct circuit structure:

```
A --------+-------------------> AND -----> X
B ----> NOT                  |
```

(Max 3 marks -- correct inputs [1], NOT gate present [1], AND gate with correct output [1])

ACCEPT: Hand-drawn or described in text

Deduct 1 mark if NOT gate is in wrong position

(c) Simplify Boolean expression [2]

(A AND B) OR (A AND B)

= A AND B [1]

- Both terms are identical [1]
- X OR X = X (idempotent law) [1]
- So the expression simplifies to A AND B [1]

(1 mark for correct answer, 1 mark for valid reasoning)

ALTERNATIVE answer: "Cannot be simplified further" = 0 marks

(The expression CAN be simplified -- it IS identical terms)

(d) Boolean expression from description [1]

X = (A AND B) OR C [1]

ACCEPT: X = C OR (A AND B)

REJECT: Missing brackets (A AND B OR C has wrong precedence)

SECTION C: SCENARIO QUESTION

QUESTION 7 (Scenario) -- 15 marks

(a)(i) Algorithm for processing a book loan [8]

Award marks for:

PROCEDURE ProcessLoan() [1]

DECLARE MemberID, ISBN, LoanDate, DueDate : STRING [1]

```
 DECLARE Valid : BOOLEAN                              [1]
 // Validate Member ID                                [1]
 Valid <-- FALSE                                      [1]
 REPEAT                                               [1]
   OUTPUT "Enter 6-digit Member ID:"                  [1]
   INPUT MemberID                                     [1]
   IF LENGTH(MemberID) = 6 THEN                       [1]
     Valid <-- TRUE                                   [1]
   ELSE                                               [1]
     OUTPUT "Invalid Member ID. Must be 6 digits."    [1]
   ENDIF                                              [1]
 UNTIL Valid = TRUE                                   [1]
 // Validate ISBN                                     [1]
 Valid <-- FALSE                                      [1]
 REPEAT                                               [1]
   OUTPUT "Enter 10-character ISBN:"                  [1]
   INPUT ISBN                                         [1]
   IF LENGTH(ISBN) = 10 THEN                          [1]
     Valid <-- TRUE                                   [1]
   ELSE                                               [1]
     OUTPUT "Invalid ISBN. Must be 10 characters."    [1]
   ENDIF                                              [1]
 UNTIL Valid = TRUE                                   [1]
 // Calculate dates                                   [1]
 LoanDate <-- GetCurrentDate()                        [1]
 DueDate <-- LoanDate + 14  // or similar logic       [1]
 // Store in file                                     [1]
 OPENFILE "Loans.txt" FOR APPEND                      [1]
 WRITEFILE "Loans.txt", MemberID & "," & ISBN & "," & LoanDate & "," & DueDate [1]
 CLOSEFILE "Loans.txt"                                [1]
 // Confirmation                                      [1]
 OUTPUT "Loan recorded successfully"                  [1]
 OUTPUT "Member:", MemberID, "ISBN:", ISBN            [1]
 OUTPUT "Due date:", DueDate                          [1]
ENDPROCEDURE                                          [1]
```

(Max 8 marks -- award best 8 points from above)

Level-based marking for 8-mark coding question:

Level 3 (7-8 marks): Complete, correct algorithm

- All inputs validated with appropriate error messages [2-3]
- Due date calculated correctly [1]
- File operations correct (open, write, close) [2]
- Confirmation message displayed [1]
- Uses appropriate subprogram structure [1]
- Well-structured with comments [1]

Level 2 (4-6 marks): Mostly correct with minor errors

- Most validations present but may be incomplete [1-2]
- File operations attempted but may have syntax errors [1-2]
- Basic structure correct [1-2]
- Some comments [1]

Level 1 (1-3 marks): Limited attempt

- Basic input/output only [1]
- Little or no validation [0-1]
- File operations not attempted or very incorrect [0-1]
- Poor structure [0-1]

Level 0 (0 marks): No relevant content

(a)(ii) How Member ID is validated [2]

- The algorithm checks the length of the input string [1]
- It uses LENGTH(MemberID) = 6 to ensure exactly 6 characters [1]
- Uses REPEAT...UNTIL to keep asking until valid input is given [1]
- Displays an error message if input is invalid [1]

(Any 2 valid points for 2 marks)

(b) Checking overdue books [3]

- Open the "Loans.txt" file for reading [1]
- Read each line and split by comma to get MemberID, ISBN, LoanDate, DueDate [1]
- Compare the member's ID with each record [1]
- Compare the DueDate with the current date [1]
- If DueDate is earlier than current date, the book is overdue [1]
- Output a list of overdue books for that member [1]

(Any 3 valid, sequential steps for 3 marks)

ACCEPT equivalent descriptions or pseudocode

(c) Limiting to 3 books [2]

- Read the Loans file and count how many books the member currently has

on loan [1]

- Before processing a new loan, check if the count is less than 3 [1]
- If count >= 3, display "Maximum books reached" and reject the loan [1]
- Only allow the loan to proceed if count < 3 [1]

(Any 2 valid points for 2 marks)

ACCEPT: Adding a counter field to the member record

ACCEPT: Checking the file for existing loans by that member

GRADE BOUNDARIES -- MOCK EXAM 3 PAPER 2

RAW MARKS (out of 73, scale to 75):

Grade	Minimum Raw Mark (/73)	Scaled Mark (/75)
A*	60	62
A	51	52
B	44	45
C	37	38
D	30	31
E	23	24
F	16	16
U	0-15	0-15

SCALING FORMULA: Scaled Mark = round((Raw Mark / 73) x 75)

Alternative (if not scaling): Use raw marks out of 73 with these boundaries:

Grade	Minimum Mark (/73)
A*	60
A	51
C	37
D	30
E	23
F	16
U	0-15

MOCK EXAM 4 -- PAPER 1 (WEEK 15)

THEORY OF COMPUTER SCIENCE

Time: 1 hour 45 minutes

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

- Write your name, class, and date in the spaces provided.
- Answer ALL questions.
- Write your answers in the spaces provided.
- If you need extra space, ask for additional paper.
- Calculators are NOT permitted.

SECTION A: SHORT ANSWER QUESTIONS (28 marks)

Answer ALL questions in this section.

QUESTION 1 (Data Representation)

(a) The ASCII code for the character 'A' is 65 in denary.

State the 8-bit binary representation of 'A'. [1]

(b) Convert the hexadecimal number 2F to denary. Show your working. [2]

(c) Explain how a check digit is used to detect errors in data entry. [2]

(d) A bitmap image has a resolution of 640 x 480 pixels and uses 256 colours.

Calculate the minimum file size of the image in kilobytes (KB).

Show your working. [3]

[Question 1 Total: 8 marks]

QUESTION 2 (Hardware and Software)

(a) State the purpose of the Control Unit (CU) in the CPU. [1]

(b) A computer has 4 GB of RAM.

(i) State the number of bytes in 4 GB. Give your answer in standard form. [2]

(ii) Explain why adding more RAM might make a computer run faster. [2]

(c) Describe what happens during the compilation of a high-level program.

Include the three main stages in your answer. [3]

[Question 2 Total: 8 marks]

QUESTION 3 (The Internet and Security)

(a) State the difference between the Internet and the World Wide Web. [2]

(b) Draw a simple diagram to show how a client computer communicates with a web server to request a web page. Label the diagram. [3]

[Draw your diagram in this space]

(c) HTTPS is used for secure communication on the web.

Explain how SSL/TLS provides security when data is transmitted. [3]

[Question 3 Total: 8 marks]

QUESTION 4 (Automated and Emerging Technologies)

(a) State two input devices that would be found in a smartphone.

For each device, state what it detects or measures. [4]

Device 1: ____

Detects: ____

Device 2: ____

Detects: ____

(b) Describe what is meant by machine learning. [2]

(c) A washing machine uses a microprocessor to control its operation.

Explain how sensors and actuators are used in a washing machine. [3]

[Question 4 Total: 9 marks]

QUESTION 5 (Data Transmission)

(a) State one advantage and one disadvantage of serial data transmission compared to parallel data transmission. [2]

Advantage: ____

Disadvantage: ____

(b) Explain how parity checking works. Your answer should include what happens when an error is detected. [3]

[Question 5 Total: 5 marks]

SECTION A TOTAL: 38 marks

SECTION B: STRUCTURED QUESTIONS (32 marks)

Answer ALL questions in this section.

QUESTION 6 (Storage and Memory)

A school needs to purchase storage devices for three different purposes:

- Storing students' examination coursework files (large documents)
- Installing the operating system and applications on new computers
- Archiving old student records that must be kept for 10 years

(a) For each purpose, recommend the most suitable type of secondary storage. Justify your choice. [6]

Purpose 1 (Coursework files): ____

Justification: ____

Purpose 2 (OS and applications): ____________________________________

Justification: ____

Purpose 3 (Archiving old records): ___________________________________

Justification: ____

(b) Explain the difference between RAM and ROM. Your answer should include the terms "volatile" and "non-volatile". [4]

(c) Describe the purpose of virtual memory. Explain one disadvantage of using it. [3]

(d) A student says: "A computer with an SSD will always boot faster than a computer with an HDD."

Evaluate this statement. [4]

[Question 6 Total: 17 marks]

QUESTION 7 (Security, Privacy and Emerging Technology)

A hospital stores patient medical records on a computer system. The records contain personal information, medical history, and current treatments.

(a) Explain why it is important that patient medical records are kept secure. Give two reasons. [3]

(b) Describe three methods the hospital could use to protect the data from unauthorised access. [6]

Method 1: ____

Method 2: ____

Method 3: ____

(c) The hospital is considering using an expert system to help doctors diagnose illnesses.

Describe two advantages and two disadvantages of using an expert system for medical diagnosis. [4]

Advantage 1: ____

Advantage 2: ____

Disadvantage 1: ____

Disadvantage 2: ____

(d) Explain how 3D printing technology could be used in a hospital.

Give one benefit and one limitation. [3]

[Question 7 Total: 16 marks]

SECTION B TOTAL: 33 marks

PAPER TOTAL: 71 marks (raw, before standardisation to 75)

Note to teacher: Standardise to 75 marks by scaling: (student mark / 71) x 75

END OF MOCK EXAM 4 -- PAPER 1

MOCK EXAM 4 -- PAPER 1: MARK SCHEME

SECTION A: SHORT ANSWER QUESTIONS

QUESTION 1 (Data Representation) -- 8 marks

(a) 8-bit binary for 'A' (ASCII 65) [1]

ACCEPT: 01000001

REJECT: 1000001 (7-bit, no mark)

(b) Hexadecimal 2F to denary [2]

Working: 2 x 16 = 32, F = 15 [1]

Answer: 32 + 15 = 47 [1]

(1 mark for method showing correct place values, 1 mark for correct answer)

(c) How check digit detects errors [2]

- A check digit is calculated from the other digits using an algorithm [1]
- When data is entered, the check digit is recalculated and compared [1]
- If the check digits do not match, an error is detected [1]
- Does not correct the error, only detects it [1]

(Any 2 valid points)

(d) Bitmap file size calculation [3]

256 colours means 8 bits per pixel (2^8 = 256) [1]

File size = 640 x 480 x 8 bits [1]

= 2,457,600 bits

= 307,200 bytes

= 300 KB [1]

(Accept 307.2 KB if student did not divide by 1024 for final step)

(Allow follow-through for arithmetic errors)

QUESTION 2 (Hardware and Software) -- 8 marks

(a) Purpose of Control Unit [1]

ACCEPT: Manages the execution of instructions

ACCEPT: Controls and coordinates the components of the CPU

ACCEPT: Sends control signals to other components

REJECT: "Controls the computer" (too vague)

(b)(i) Number of bytes in 4 GB [2]

4 GB = 4 x 1024 x 1024 x 1024 bytes [1]

= 4,294,967,296 bytes

= 4.29 x 10^9 bytes (accept 4.3 x 10^9) [1]

(1 mark for correct calculation method, 1 mark for standard form)

(b)(ii) Why more RAM makes computer faster [2]

- More RAM means more programs/data can be held in memory at once [1]
- Less need to use virtual memory (hard disk as RAM) [1]
- Virtual memory is much slower than RAM [1]
- Programs can run without swapping data to disk [1]

(Any 2 valid points)

(c) Stages of compilation [3]

Award 1 mark each for:

- Lexical analysis: source code converted to tokens/symbols [1]
- Syntax analysis: tokens checked against grammar rules / parse tree created [1]
- Code generation: machine code produced [1]
- (Accept "code optimisation" as additional stage, but max 3 marks)

(Must be in correct order for full marks)

QUESTION 3 (The Internet and Security) -- 8 marks

(a) Internet vs World Wide Web [2]

- The Internet is a global network of interconnected computers [1]
- The World Wide Web is a system of interlinked hypertext documents

accessed via the Internet [1]

- The WWW is a SERVICE that runs ON the Internet [1]
- The Internet includes email, FTP, etc. -- not just the Web [1]

(Any 2 clear distinctions)

(b) Client-server diagram [3]

Award marks for:

- Client computer shown on left [1]
- Web server shown on right [1]
- Internet/cloud/network between them [1]
- Labels on client, server, and connection [1]
- Arrow showing request from client to server [1]
- Arrow showing response from server to client [1]

(Max 3 marks -- 1 for client, 1 for server, 1 for correct connection)

ACCEPT: Described in text if diagram unclear

(c) How SSL/TLS provides security [3]

- SSL/TLS encrypts data transmitted between client and server [1]
- Uses public and private keys for encryption/decryption [1]
- Data is scrambled so it cannot be read if intercepted [1]
- Provides authentication (confirms server identity) [1]
- Creates a secure "tunnel" for data transmission [1]
- Indicated by HTTPS and padlock icon in browser [1]

(Any 3 valid points)

QUESTION 4 (Automated and Emerging Technologies) -- 9 marks

(a) Two smartphone input devices [4]

Award 1 mark for device + 1 mark for what it detects (x2):

- Touchscreen / capacitive sensor [1] -- detects finger position/pressure [1]
- Accelerometer [1] -- detects motion/orientation [1]
- Gyroscope [1] -- detects rotation/angular velocity [1]
- GPS receiver [1] -- detects geographical location [1]
- Proximity sensor [1] -- detects distance to objects [1]
- Ambient light sensor [1] -- detects light levels [1]
- Fingerprint sensor [1] -- detects biometric data [1]
- Microphone [1] -- detects sound [1]
- Camera [1] -- detects light/images [1]

(Any 2 valid pairs, max 4 marks)

(b) Machine learning [2]

- A type of AI where computer systems learn from data [1]
- Systems improve performance on a task through experience [1]
- Without being explicitly programmed for every scenario [1]
- Uses algorithms to find patterns in data [1]

(Any 2 valid points)

(c) Sensors and actuators in washing machine [3]

- Sensors detect conditions (temperature sensor for water temperature, pressure sensor for water level) [1]

- The microprocessor reads sensor data and makes decisions [1]
- Actuators carry out physical actions (motor to rotate drum, valve to let water in/out, heater to warm water) [1]
- The system uses feedback: sensors check if conditions are correct, actuators adjust accordingly [1]

(Any 3 valid points)

QUESTION 5 (Data Transmission) -- 5 marks

(a) Serial vs parallel [2]

Advantage of serial:

- Cheaper (fewer wires needed) [1]
- Works over longer distances [1]
- Less interference/crosstalk [1]

Disadvantage of serial:

- Slower data transfer rate than parallel [1]
- Data must be sent one bit at a time [1]

(1 mark for advantage, 1 mark for disadvantage)

(b) How parity checking works [3]

- An extra parity bit is added to make the total number of 1-bits odd (odd parity) or even (even parity) [1]
- When data is received, the parity is checked [1]
- If the parity is wrong (unexpected number of 1-bits), an error has been detected [1]
- The receiver requests retransmission (ARQ) or alerts the user [1]
- Parity checking can detect single-bit errors but not all errors [1]

(Any 3 valid points)

SECTION B: STRUCTURED QUESTIONS

QUESTION 6 (Storage and Memory) -- 17 marks

(a) Storage recommendations [6]

Purpose 1 (Coursework files):

- HDD or SSD or Cloud storage [1]
- Justification: Large files need high capacity [1]
- HDD offers large capacity at lower cost / SSD is faster [1]
- Cloud allows access from anywhere and automatic backup [1]

(Storage type [1] + valid justification [1])

Purpose 2 (OS and applications):

- SSD [1]
- Justification: Fast read/write speeds mean faster booting [1]
- No moving parts so more reliable / durable [1]
- OS needs to be accessed frequently, speed is important [1]

(SSD [1] + valid justification [1])

Purpose 3 (Archiving old records):

- Optical storage (DVD/Blu-ray) or HDD (external) or Tape [1]
- Justification: Long-term storage, data integrity preserved [1]
- Optical is not easily overwritten / tape is cheap for bulk [1]
- Kept off-site for disaster recovery [1]

(Storage type [1] + valid justification [1])

(b) RAM vs ROM [4]

- RAM is volatile (loses contents when power is turned off) [1]
- ROM is non-volatile (retains contents when power is turned off) [1]
- RAM can be read from and written to (read/write) [1]
- ROM can only be read from (read-only) [1]
- RAM stores currently running programs and data [1]
- ROM stores boot instructions / firmware [1]
- RAM has larger capacity than ROM [1]

(Need both "volatile" and "non-volatile" mentioned for full marks,

plus any 2 other valid differences. Max 4 marks)

(c) Virtual memory [3]

- Virtual memory uses part of the hard disk as an extension of RAM [1]
- When RAM is full, less-used data is moved to the hard disk [1]
- Allows more programs to run than physical RAM would allow [1]

Disadvantage:

- Much slower than RAM (hard disk access is slow) [1]
- Can cause the computer to slow down significantly [1]
- Frequent disk access wears out HDD/SSD [1]
- Called "thrashing" when excessive [1]

(Virtual memory description [2] + disadvantage [1])

(d) Evaluate statement about SSD booting [4]

Award marks using levels approach:

Level 2 (3-4 marks): Balanced evaluation

- SSDs generally DO boot faster than HDDs [1]
- SSDs have no moving parts / use flash memory = faster random access [1]
- Data can be read instantly from any location [1]
- BUT boot speed also depends on other factors: RAM amount, number of

startup programs, CPU speed [1]

- If an SSD is nearly full or is a low-quality model, it may be slower [1]
- Conclusion: Statement is generally true but not ALWAYS true [1]

(Max 4 marks with balanced view and conclusion)

Level 1 (1-2 marks): Basic response

- Agrees that SSD is faster [1]

- Mentions no moving parts [1]
- No consideration of other factors [1]

Level 0 (0 marks): No relevant content

QUESTION 7 (Security, Privacy and Emerging Technology) -- 16 marks

(a) Why medical records must be secure [3]

- Contains sensitive personal information that must be kept private [1]
- Legal requirement (data protection laws) [1]
- If accessed by unauthorised persons, patients could be harmed [1]
- Information could be used for identity theft or fraud [1]
- Patients trust the hospital to keep their data confidential [1]

(Any 3 valid points)

(b) Three security methods [6]

Award 1 mark for method + 1 mark for description (x3):

- Encryption [1]: Data is scrambled so only authorised users with the

key can read it [1]

- Access control / Authentication [1]: Passwords, biometrics, or

smart cards to verify identity [1]

- Firewall [1]: Filters network traffic to block unauthorised access [1]
- Audit trail / Access logs [1]: Records who accessed what data and when [1]
- Physical security [1]: Locked rooms, restricted access to servers [1]
- Staff training [1]: Educating staff about security threats [1]
- Regular backups [1]: Ensuring data can be recovered if lost [1]

(Any 3 valid method + description pairs)

(c) Expert system advantages and disadvantages [4]

Advantages (max 2):

- Available 24/7 (does not tire or need breaks) [1]
- Can store knowledge of many experts in one system [1]
- Consistent recommendations (not affected by emotions) [1]
- Can help in remote areas where specialists are not available [1]
- Faster initial diagnosis [1]

Disadvantages (max 2):

- Lacks human judgement and empathy [1]
- Cannot handle unusual cases not in the knowledge base [1]
- Expensive to develop and maintain [1]
- Doctors may become overly dependent on the system [1]
- Responsibility if the system makes a wrong diagnosis [1]
- Needs constant updating with latest medical knowledge [1]

(2 advantages [2] + 2 disadvantages [2])

(d) 3D printing in hospitals [3]

- Can be used to create prosthetic limbs, surgical models, or implants [1]

- Custom-made to fit individual patients [1]
- Benefit: Reduces surgery time / allows pre-surgical practice [1]
- Benefit: Customised healthcare / faster production [1]
- Limitation: Expensive equipment and materials [1]
- Limitation: Limited materials suitable for medical use [1]
- Limitation: Slow production for complex items [1]

(Use [1] + benefit [1] + limitation [1])

GRADE BOUNDARIES -- MOCK EXAM 4 PAPER 1

RAW MARKS (out of 71, scale to 75):

Grade	Minimum Raw Mark (/71)	Scaled Mark (/75)
A*	58	61
A	50	53
B	43	45
C	36	38
D	29	31
E	22	23
F	15	16
U	0-14	0-14

SCALING FORMULA: Scaled Mark = round((Raw Mark / 71) x 75)

MOCK EXAM 5 -- PAPER 2 (WEEK 16)

PROBLEM-SOLVING AND PROGRAMMING

Time: 1 hour 45 minutes

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

- Write your name, class, and date in the spaces provided.
- Answer ALL questions.
- Write your answers in the spaces provided.
- Calculators are NOT permitted.

PSEUDOCODE REFERENCE (provided in exam)

DECLARE <variable> : <type>

CONSTANT <name> = <value>

<identifier> <-- <value>

INPUT <identifier>

OUTPUT <identifier/value/string>

IF <condition> THEN … ELSE … ENDIF

CASE OF <identifier>:

<value>: <statement(s)>

OTHERWISE: <statement(s)>

ENDCASE

FOR <identifier> <-- <value1> TO <value2> STEP <value> … NEXT

REPEAT … UNTIL <condition>

WHILE <condition> DO … ENDWHILE

FUNCTION <name>(<parameters>) RETURNS <type> … ENDFUNCTION

PROCEDURE <name>(<parameters>) … ENDPROCEDURE

RETURN <value>

CALL <name>(<parameters>)

COMMENT // <text>

AND, OR, NOT

LENGTH(<string>)

SUBSTRING(<string>, <start>, <length>)

ASC(<character>)

RANDOM()

ROUND(<number>, <decimal places>)

MOD, DIV

SECTION A: SHORT ANSWER QUESTIONS (24 marks)

Answer ALL questions in this section.

QUESTION 1 (Algorithm Design and Problem Solving)

(a) State two benefits of using subprograms (functions or procedures) in a program. [2]

7. ____
8. ____

(b) Explain what is meant by the term "parameter" in programming. [2]

(c) Describe the difference between count-controlled and condition-controlled iteration. Give an example of when each would be used. [4]

[Question 1 Total: 8 marks]

QUESTION 2 (Programming Constructs and String Handling)

(a) Look at this pseudocode:

```
DECLARE Count : INTEGER
Count <-- 1
WHILE Count <= 5 DO
  OUTPUT SUBSTRING("COMPUTER", Count, 1)
  Count <-- Count + 1
ENDWHILE
```

(i) Trace through the code and write the output produced. [2]

(ii) State what would happen if SUBSTRING("COMPUTER", Count, 1) was changed to SUBSTRING("COMPUTER", Count, 2). [1]

(b) Write a pseudocode statement to calculate the result of 17 divided by 5, giving only the whole number part (no remainder). [1]

(c) Look at this pseudocode:

```
DECLARE Score : INTEGER
INPUT Score
CASE OF Score
  90 TO 100: OUTPUT "A"
  80 TO 89:  OUTPUT "B"
  70 TO 79:  OUTPUT "C"
  60 TO 69:  OUTPUT "D"
  OTHERWISE: OUTPUT "F"
ENDCASE
```

(i) State the output when the input is 85. [1]

(ii) State the output when the input is 55. [1]

(iii) Explain one advantage of using a CASE statement instead of multiple nested IF statements for this problem. [2]

(d) Explain why it is good practice to use meaningful variable names rather than single letters when writing programs. [2]

[Question 2 Total: 9 marks]

QUESTION 3 (Databases, SQL and Boolean Logic)

A shop uses a database to manage its products. The database has one table:

PRODUCT (ProductID, ProductName, Category, Price, StockLevel, Supplier)

(a) State the purpose of a primary key in a database table. [1]

(b) Explain the difference between a field and a record in a database. [2]

(c) Write an SQL statement to display all products in the "Electronics" category, sorted by price from highest to lowest. [3]

(d) Write an SQL statement to find the average price of all products. [2]

(e) Complete the truth table for the Boolean expression:

X = (A AND B) OR (NOT C)

A	B	C	A AND B	NOT C	X = (A AND B) OR (NOT C)
0	0	1			
0	1	0			
0	1	1			
1	0	0			

| 1 | 0 | 1 | | | |

| 1 | 1 | 0 | | | |

[4]

[Question 3 Total: 12 marks]

SECTION A TOTAL: 29 marks

SECTION B: CODE READING AND WRITING (31 marks)

Answer ALL questions in this section.

QUESTION 4 (Trace Tables and Algorithm Analysis)

Study the following pseudocode:

```
DECLARE Numbers : ARRAY[1:8] OF INTEGER
DECLARE i, j, Temp : INTEGER
Numbers[1] <-- 64
Numbers[2] <-- 25
Numbers[3] <-- 12
Numbers[4] <-- 88
Numbers[5] <-- 5
Numbers[6] <-- 37
Numbers[7] <-- 91
Numbers[8] <-- 18
FOR i <-- 1 TO 7
 FOR j <-- 1 TO 8 - i
  IF Numbers[j] > Numbers[j + 1] THEN
   Temp <-- Numbers[j]
   Numbers[j] <-- Numbers[j + 1]
   Numbers[j + 1] <-- Temp
  ENDIF
 NEXT
NEXT
FOR i <-- 1 TO 8
 OUTPUT Numbers[i]
NEXT
```

(a) Name the sorting algorithm shown in this pseudocode. [1]

(b) Complete the trace table showing the state of the Numbers array after

each pass of the outer loop (i = 1, i = 2, i = 3):

Pass | Numbers[1] | Numbers[2] | Numbers[3] | Numbers[4] | Numbers[5] | Numbers[6] | Numbers[7] | Numbers[8]

Initial	64	25	12	88	5	37	91	18
i=1								
i=2								
i=3								

[6]

(c) State the final output of the algorithm (all 8 numbers in order). [1]

(d) State the purpose of the variable Temp in this algorithm. [1]

[Question 4 Total: 9 marks]

QUESTION 5 (Pseudocode Writing -- Arrays and File Handling)

(a) A teacher wants to record test scores for a class of 30 students.

Write pseudocode to:

- Declare an array called Scores to hold 30 integer values
- Use a loop to input each student's score (between 0 and 100)
- Validate that each score is between 0 and 100
- Count how many students scored 50 or above
- Output the count

You should declare all variables used. [7]

(b) Write pseudocode to read a text file called "Students.txt" that contains one student name per line. Your algorithm should:

- Open the file for reading
- Count how many lines (students) are in the file
- Display each name as it is read
- Close the file
- Output the total number of students

You should declare all variables used. [5]

(c) Look at this pseudocode:

```
FUNCTION IsEven(Num : INTEGER) RETURNS BOOLEAN
  IF Num MOD 2 = 0 THEN
    RETURN TRUE
  ELSE
    RETURN FALSE
  ENDIF
ENDFUNCTION
DECLARE x : INTEGER
x <-- 15
IF IsEven(x) THEN
  OUTPUT x, "is even"
ELSE
```

```
  OUTPUT x, "is odd"
ENDIF
```

(i) State the output of this pseudocode. [1]

(ii) State one advantage of using a function like IsEven instead of writing the MOD check directly in the main program. [2]

[Question 5 Total: 15 marks]

QUESTION 6 (Boolean Logic Circuits)

(a) Draw the logic circuit for the Boolean expression:

X = (NOT A) OR (B AND C)

Label all inputs and the output. [3]

[Draw in this space]

(b) Complete the truth table for the expression in part (a):

A	B	C	NOT A	B AND C	X = (NOT A) OR (B AND C)
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			

[3]

(c) Simplify the Boolean expression below using Boolean algebra rules.

Show your working. [2]

A AND (A OR B)

(d) A lift (elevator) door control system has three inputs:

- M: Motor running (1 = running, 0 = stopped)
- B: Button pressed (1 = button pressed, 0 = not pressed)
- S: Safety sensor clear (1 = clear, 0 = obstruction detected)

The door opens (output D = 1) when:

- The motor is stopped AND the button is pressed, OR
- The safety sensor detects an obstruction

Write the Boolean expression for D. [1]

D = ____

[Question 6 Total: 9 marks]

SECTION B TOTAL: 33 marks

SECTION C: SCENARIO QUESTION (15 marks)

Answer this question in the spaces provided.

QUESTION 7 (Scenario)

A teacher needs a program to manage a multiple-choice quiz for students.

The quiz has 10 questions. Each question has 4 possible answers (A, B, C, D).

Only one answer is correct for each question.

The program must:

- Store the 10 correct answers in an array called CorrectAnswers
- Allow a student to enter their name
- Ask the student each of the 10 questions and record their answer
- Validate that the student's answer is A, B, C, or D (reject other inputs)
- Compare each student answer with the correct answer
- Calculate and display the student's score (out of 10)
- Display "PASS" if score is 7 or above, otherwise display "RETRY"
- Store the student's name and score in a file called "Results.txt"

The correct answers are: A, C, B, D, A, B, C, D, A, C

(a) (i) Write an algorithm in pseudocode to run this quiz program.

Your algorithm should meet all the requirements listed above.

You may assume the questions are displayed separately and the student only enters their answer choice (A, B, C, or D).

You should use modular design (at least one function or procedure). [10]

(ii) Explain how your algorithm validates the student's answer. [2]

(b) The teacher wants to find the highest score from all students who took the quiz. The scores are stored in "Results.txt" with each line containing a student name and score separated by a comma.

Describe how a program could find and display the highest score and the name of the student who achieved it. You do not need to write pseudocode. [3]

[Question 7 Total: 15 marks]

SECTION C TOTAL: 15 marks

PAPER TOTAL: 77 marks (raw, before standardisation to 75)

Note to teacher: Standardise to 75 marks by scaling: (student mark / 77) x 75

END OF MOCK EXAM 5 -- PAPER 2

MOCK EXAM 5 -- PAPER 2: MARK SCHEME

SECTION A: SHORT ANSWER QUESTIONS

QUESTION 1 (Algorithm Design and Problem Solving) -- 8 marks

(a) Benefits of subprograms [2]

Award 1 mark each for any 2 of:

- Code can be reused multiple times without rewriting [1]
- Programs are easier to read and understand (modular) [1]
- Easier to test and debug (test each part separately) [1]

- Different programmers can work on different subprograms [1]
- Changes only need to be made in one place [1]

(b) Parameter [2]

- A value that is passed into a subprogram (function or procedure) [1]
- When the subprogram is called, the parameter is given a value (argument) [1]
- Allows the same subprogram to work with different data each time [1]
- Acts as a local variable within the subprogram [1]

(Any 2 valid points)

(c) Count-controlled vs condition-controlled iteration [4]

Count-controlled:

- The number of repetitions is known before the loop starts [1]
- Uses a counter (e.g., FOR loop) [1]
- Example: Processing exactly 100 records [1]

Condition-controlled:

- The loop continues until a condition is met (unknown number of repetitions) [1]
- Uses a condition (e.g., WHILE or REPEAT...UNTIL) [1]
- Example: Reading data until end of file [1]
- Example: Input validation until correct input received [1]

(2 marks for each type: description + example)

QUESTION 2 (Programming Constructs and String Handling) -- 9 marks

(a)(i) Output of SUBSTRING loop [2]

ACCEPT: C, O, M, P, U (or C O M P U or COMP)

ACCEPT: One character per line

REJECT: "COMPUTER" (did not trace correctly)

Working: SUBSTRING("COMPUTER", Count, 1) extracts 1 character starting

at position Count. Loop runs Count = 1 to 5, extracting positions 1-5.

(1 mark for correct understanding, 1 mark for correct 5 characters)

(a)(ii) Effect of changing length to 2 [1]

ACCEPT: Would output 2 characters at a time (CO, MP, UT, ER, and error)

ACCEPT: Would extract 2 characters instead of 1

ACCEPT: Output would be "CO", "MP", "UT", "ER" then out of range

(b) DIV statement [1]

ACCEPT: Result <-- 17 DIV 5

ACCEPT: OUTPUT 17 DIV 5

ACCEPT: Any valid assignment using DIV operator

Answer should give 3

(c)(i) Output for input 85 [1]

ACCEPT: B

(c)(ii) Output for input 55 [1]

ACCEPT: F

(c)(iii) Advantage of CASE over nested IF [2]

- CASE is easier to read and understand [1]
- CASE is more efficient (faster execution) [1]
- CASE avoids deeply nested structures [1]
- Easier to add or remove options [1]
- Less chance of logic errors [1]

(Any 2 valid points)

(d) Meaningful variable names [2]

- Makes the code easier to understand for other programmers [1]
- Easier to maintain and debug (purpose of variable is clear) [1]
- Self-documenting code (reduces need for comments) [1]
- Reduces errors from confusing similar variables [1]

(Any 2 valid points)

QUESTION 3 (Databases, SQL and Boolean Logic) -- 12 marks

(a) Purpose of primary key [1]

ACCEPT: Uniquely identifies each record in the table

ACCEPT: Ensures no two records have the same key value

ACCEPT: Provides a way to identify a specific record

REJECT: "Identifies the table" (confused with table name)

(b) Field vs record [2]

- A field is a single column in a table / one piece of information [1]
- A record is a complete row in a table / all fields for one item [1]
- Example: In a student table, "FirstName" is a field, one student's

complete details form a record [1]

(1 mark for each correct definition)

(c) SQL for Electronics category sorted by price [3]

SELECT * [1]

FROM PRODUCT [1]

WHERE Category = 'Electronics' [1]

ORDER BY Price DESC [1]

(Max 3 marks -- deduct 1 if DESC missing, still give 2)

ACCEPT: ProductName, Price (specific fields instead of *)

ACCEPT: "Electronics" with double quotes

REJECT: Missing quotes around Electronics (0 for WHERE clause)

(d) SQL for average price [2]

SELECT AVG(Price) [1]

FROM PRODUCT [1]

ACCEPT: SELECT AVG(Price) AS AveragePrice

ACCEPT: AVERAGE instead of AVG (minor error, 1 mark)

REJECT: Missing FROM clause (max 1 mark)

(e) Truth table for X = (A AND B) OR (NOT C) [4]

A	B	C	A AND B	NOT C	X = (A AND B) OR (NOT C)
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	0	1	1
1	1	1	1	0	1

Award 1 mark for correct A AND B column

Award 1 mark for correct NOT C column

Award 2 marks for correct X column (all 8 rows)

(Deduct 1 mark per error in X column, minimum 0)

SECTION B: CODE READING AND WRITING

QUESTION 4 (Trace Tables and Algorithm Analysis) -- 9 marks

(a) Name of sorting algorithm [1]

ACCEPT: Bubble sort

ACCEPT: Bubble Sort

REJECT: Sorting algorithm (too vague)

(b) Trace table after each pass [6]

Pass	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	
Initial	64	25	12	88	5	37	91	18	
i=1	25	12	64	5	37	88	18	91	[2]
i=2	12	25	5	37	64	18	88	91	[2]
i=3	12	5	25	37	18	64	88	91	[2]

Award marks per row:

i=1 row correct: [2] (all 8 positions correct)

i=1 row mostly correct (6-7 correct): [1]

i=2 row correct: [2]

i=2 row mostly correct: [1]

i=3 row correct: [2]

i=3 row mostly correct: [1]

(Allow follow-through from previous row if only one number out of place)

(Bubble sort: largest element "bubbles" to the end each pass)

(c) Final output [1]

ACCEPT: 5, 12, 18, 25, 37, 64, 88, 91

ACCEPT: Any format showing sorted order

(Allow follow-through from trace table)

(d) Purpose of Temp [1]

ACCEPT: To temporarily hold a value during swapping

ACCEPT: Used to swap two values in the array

ACCEPT: Temporary storage during exchange

REJECT: "Stores a number" (too vague)

QUESTION 5 (Pseudocode Writing -- Arrays and File Handling) -- 15 marks

(a) Test scores pseudocode [7]

Award marks for:

```
DECLARE Scores : ARRAY[1:30] OF INTEGER          [1]
DECLARE i, Count, Score : INTEGER                [1]
Count <-- 0                                      [1]
FOR i <-- 1 TO 30                                [1]
 REPEAT                                          [1]
  OUTPUT "Enter score for student", i            [1]
  INPUT Score                                    [1]
  IF Score < 0 OR Score > 100 THEN               [1]
   OUTPUT "Invalid score. Must be 0 to 100."     [1]
  ENDIF                                          [1]
 UNTIL Score >= 0 AND Score <= 100               [1]
 Scores[i] <-- Score                             [1]
 IF Score >= 50 THEN                             [1]
  Count <-- Count + 1                            [1]
 ENDIF                                           [1]
NEXT                                             [1]
OUTPUT "Number of students scoring 50 or above:", Count    [1]
```

(Max 7 marks -- award best 7 points)

Key marking points:

- Correct array declaration [1]
- Loop through all 30 students [1]

- Input validation (0 to 100) [2]
- Store score in array [1]
- Count scores >= 50 [2]
- Output the count [1]
- Proper structure (ENDIF, NEXT, etc.) [1]

(b) File reading pseudocode [5]

Award marks for:

DECLARE StudentName : STRING [1]

DECLARE Count : INTEGER [1]

Count <-- 0 [1]

OPENFILE "Students.txt" FOR READ [1]

WHILE NOT EOF("Students.txt") DO [1]

READFILE "Students.txt", StudentName [1]

OUTPUT StudentName [1]

Count <-- Count + 1 [1]

ENDWHILE [1]

CLOSEFILE "Students.txt" [1]

OUTPUT "Total number of students:", Count [1]

(Max 5 marks -- award best 5 points)

Key marking points:

- Open file for reading [1]
- Loop until EOF [1]
- Read each line [1]
- Display name [1]
- Count students [1]
- Close file [1]
- Output total [1]

ALTERNATIVE using REPEAT...UNTIL EOF: accept if correct

(c)(i) Output of IsEven function [1]

ACCEPT: "15 is odd"

ACCEPT: "15 is odd" (any format showing x and "odd")

(Follow-through from correct understanding of the function)

(c)(ii) Advantage of using IsEven function [2]

- The code is reusable -- can check any number for evenness [1]
- The main program is easier to read and understand [1]
- The logic is in one place -- easy to change if needed [1]
- Easier to test (can test the function separately) [1]

(Any 2 valid points)

QUESTION 6 (Boolean Logic Circuits) -- 9 marks

(a) Logic circuit for X = (NOT A) OR (B AND C) [3]

Award marks for:

- Input A connected to NOT gate [1]
- Inputs B and C connected to AND gate [1]
- NOT output and AND output connected to OR gate [1]
- OR gate output labelled X [1]

Correct structure:

```
A -----> NOT -----+
                  +---> OR -----> X
B ------+         |      |
        +---> AND-+      |
C ------+                |
```

(Max 3 marks)

(b) Truth table [3]

A	B	C	NOT A	B AND C	X = (NOT A) OR (B AND C)
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	0	1	1

Award 1 mark for correct NOT A column

Award 1 mark for correct B AND C column

Award 1 mark for correct X column (allow follow-through)

(c) Simplify A AND (A OR B) [2]

Using absorption law: A AND (A OR B) = A [1]

- If A is 0, the result is 0
- If A is 1, (1 OR B) = 1, so 1 AND 1 = 1
- Therefore the result depends only on A [1]
- The B term is absorbed [1]

(1 mark for correct answer A, 1 mark for valid reasoning)

(d) Boolean expression for lift door [1]

D = (NOT M AND B) OR (NOT S) [1]

ACCEPT: D = (M=0 AND B=1) OR (S=0)

ACCEPT: D = (NOT M AND B) OR NOT S

REJECT: Wrong logic (must match description)

SECTION C: SCENARIO QUESTION

QUESTION 7 (Scenario) -- 15 marks

(a)(i) Quiz program pseudocode [10]

Award marks for:

```
PROCEDURE RunQuiz()                                          [1]
  DECLARE CorrectAnswers : ARRAY[1:10] OF STRING          [1]
  DECLARE StudentAnswers : ARRAY[1:10] OF STRING          [1]
  DECLARE Name : STRING                                 [1]
  DECLARE Score, i : INTEGER                            [1]
  DECLARE Answer : STRING                               [1]
  // Set up correct answers                             [1]
  CorrectAnswers[1] <-- "A"
  CorrectAnswers[2] <-- "C"
  CorrectAnswers[3] <-- "B"
  CorrectAnswers[4] <-- "D"
  CorrectAnswers[5] <-- "A"
  CorrectAnswers[6] <-- "B"
  CorrectAnswers[7] <-- "C"
  CorrectAnswers[8] <-- "D"
  CorrectAnswers[9] <-- "A"
  CorrectAnswers[10] <-- "C"                            [1]
  // Get student name                                   [1]
  OUTPUT "Enter your name:"
  INPUT Name
  // Ask questions                                      [1]
  Score <-- 0                                           [1]
  FOR i <-- 1 TO 10                                     [1]
    REPEAT                                              [1]
      OUTPUT "Question", i, ". Enter your answer (A/B/C/D):"
      INPUT Answer                                      [1]
      IF Answer <> "A" AND Answer <> "B" AND Answer <> "C"
        AND Answer <> "D" THEN                          [1]
       OUTPUT "Invalid answer. Please enter A, B, C, or D."[1]
      ENDIF                                             [1]
    UNTIL Answer = "A" OR Answer = "B" OR Answer = "C"
```

```
      OR Answer = "D"                    [1]
   StudentAnswers[i] <-- Answer                   [1]
   IF Answer = CorrectAnswers[i] THEN             [1]
    Score <-- Score + 1                   [1]
   ENDIF                              [1]
  NEXT                                [1]
  // Display results                      [1]
  OUTPUT "Quiz complete!"
  OUTPUT Name, "scored", Score, "out of 10"           [1]
  IF Score >= 7 THEN                      [1]
   OUTPUT "PASS"                        [1]
  ELSE                                [1]
   OUTPUT "RETRY"                       [1]
  ENDIF                               [1]
  // Save to file                         [1]
  OPENFILE "Results.txt" FOR APPEND               [1]
  WRITEFILE "Results.txt", Name & "," & Score         [1]
  CLOSEFILE "Results.txt"                   [1]
 ENDPROCEDURE                             [1]
```

(Max 10 marks -- award best 10 points)

Level-based marking for 10-mark coding question:

Level 3 (8-10 marks): Complete, correct algorithm

- CorrectAnswers array initialised correctly [1]
- Student name input [1]
- Loop through 10 questions [1]
- Input validation for A/B/C/D [2]
- Comparison with correct answers [1]
- Score calculated correctly [1]
- PASS/RETRY output based on score [1]
- Results saved to file correctly [2]
- Modular design used [1]
- Good structure and comments [1]

Level 2 (4-7 marks): Mostly correct with some errors

- Most elements present but some missing or incorrect [3-5]
- Validation attempted but incomplete [1-2]
- File operations attempted [1-2]
- Basic structure correct [1]

Level 1 (1-3 marks): Limited attempt

- Basic input/output only [1-2]
- Little or no validation [0-1]
- No file operations [0]
- Poor structure [0-1]

Level 0 (0 marks): No relevant content

(a)(ii) How answer is validated [2]

- Uses REPEAT...UNTIL to keep asking until valid input received [1]
- Checks answer is one of "A", "B", "C", or "D" [1]
- Displays error message for invalid input [1]
- Uses AND to check all invalid options / OR to check valid options [1]

(Any 2 valid points)

(b) Finding highest score [3]

- Open "Results.txt" for reading [1]
- Read each line, split by comma to get name and score [1]
- Keep track of highest score found so far and the student name [1]
- Compare each score with current highest [1]
- If higher, update highest score and student name [1]
- After reading all lines, display the name and highest score [1]
- Close the file [1]

(Any 3 valid, sequential steps for 3 marks)

GRADE BOUNDARIES -- MOCK EXAM 5 PAPER 2

RAW MARKS (out of 77, scale to 75):

Grade	Minimum Raw Mark (/77)	Scaled Mark (/75)
A*	63	61
A	54	53
B	47	46
C	39	38
D	31	30
E	24	23
F	16	16
U	0-15	0-14

SCALING FORMULA: Scaled Mark = round((Raw Mark / 77) x 75)

PART 3: INDIVIDUAL GAP ANALYSIS TEMPLATES

TEMPLATE 1: PAPER 1 PROGRESS TRACKER (MOCK 1 --> MOCK 2 --> MOCK 4)

STUDENT NAME: ____________________________ CLASS: __________

Instructions: For each topic, record the marks scored out of the maximum available in each mock exam. Shade the cell green (improved), red (declined), or yellow (same).

SYLLABUS TOPIC (Paper 1)	Max	Mock 1	Mock 2	Mock 4	Trend
1.1 Binary & Hexadecimal	8	__/8	__/8	__/8	
1.2 Data Storage (images, sound)	8	__/8	__/8	__/8	
1.3 Data Transmission	6	__/6	__/6	__/6	
1.4 Hardware (CPU, memory, FDE)	10	__/10	__/10	__/10	
1.5 Software (OS, translators)	8	__/8	__/8	__/8	
2.1 Networks (topologies, IP)	8	__/8	__/8	__/8	
2.2 Internet technologies	8	__/8	__/8	__/8	
2.3 Security (threats, cookies)	8	__/8	__/8	__/8	
3.1 Automated systems	8	__/8	__/8	__/8	
3.2 AI and Expert Systems	7	__/7	__/7	__/7	
3.3 Emerging technologies	4	__/4	__/4	__/4	
PAPER 1 TOTAL	75	__/75	__/75	__/75	

MY TOP 3 WEAKEST TOPICS (Paper 1):

9. ____ Score: ___/___
10. ____ Score: ___/___
11. ____ Score: ___/___

MY STRONGEST TOPIC (Paper 1):

____ Score: ___/___

TEACHER COMMENTS:

TEMPLATE 2: PAPER 2 PROGRESS TRACKER (MOCK 1 --> MOCK 3 --> MOCK 5)

STUDENT NAME: ___________________________ CLASS: __________

SYLLABUS TOPIC (Paper 2)	Max	Mock 1	Mock 3	Mock 5	Trend
1.1 Algorithm design	8	__/8	__/8	__/8	
1.2 Pseudocode (selection, loops)	10	__/10	__/10	__/10	
1.3 Arrays and file handling	10	__/10	__/10	__/10	
1.4 Subprograms (functions, proc)	10	__/10	__/10	__/10	
1.5 Code reading & trace tables	12	__/12	__/12	__/12	
2.1 Database concepts	5	__/5	__/5	__/5	
2.2 SQL queries	8	__/8	__/8	__/8	
2.3 Boolean logic & circuits	10	__/10	__/10	__/10	
Scenario question (15 marks)	15	__/15	__/15	__/15	
PAPER 2 TOTAL	75	__/75	__/75	__/75	

MY TOP 3 WEAKEST TOPICS (Paper 2):

12. ____ Score: ___/___
13. ____ Score: ___/___
14. ____ Score: ___/___

MY STRONGEST TOPIC (Paper 2):

____ Score: ___/___

TEACHER COMMENTS:

TEMPLATE 3: TOP 3 WEAK TOPICS IDENTIFICATION (PER STUDENT)

STUDENT NAME: ______________________________ CLASS: __________ DATE: _____________

STEP 1: List all topics where you scored LESS THAN 50%:

#	Topic	Your Score	Max	Percentage
1				

2				
3				
4				
5				
6				
7				
8				

STEP 2: Select your TOP 3 priority topics (the ones you need to revise most):

Priority 1: ____

Why this needs work: _______________________________________

What I will do to improve: ___________________________________

Target score for next mock: ___/___

Priority 2: ____

Why this needs work: _______________________________________

What I will do to improve: ___________________________________

Target score for next mock: ___/___

Priority 3: ____

Why this needs work: _______________________________________

What I will do to improve: ___________________________________

Target score for next mock: ___/___

STEP 3: Revision plan

Week	Topic to revise	Resources to use	Time planned
Week 11			
Week 12			
Week 13			
Week 14			

TEACHER SIGN-OFF: _____________________ DATE: _________

TEMPLATE 4: CLASS-LEVEL ANALYSIS TEMPLATE

CLASS: ______________ ANALYSED BY: ______________ DATE: _________

For each topic, count how many students scored LESS THAN 50%.

If more than 50% of students struggled, mark as CLASS PRIORITY.

Topic	# Students	# Scored <50%	% Struggling	Priority?
Binary & Hexadecimal				
Data Storage				
Data Transmission				
Hardware (CPU,				

FDE)				
Software (OS, Translators)				
Networks				
Internet Technologies				
Security				
Automated Systems				
AI and Expert Systems				
Algorithms				
Programming (selection, loops)				
Arrays and File Handling				
Subprograms				
Trace Tables				
Databases & SQL				
Boolean Logic				
Scenario Questions				

CLASS PRIORITY TOPICS (whole-class revision needed):

15. ____
16. ____
17. ____

INDIVIDUALS NEEDING EXTRA SUPPORT:

Name: ______________________ Weak areas: _____________________________________

Name: ______________________ Weak areas: _____________________________________

Name: ______________________ Weak areas: _____________________________________

PLANNED INTERVENTIONS:

PART 4: TARGET SETTING SHEETS

TEMPLATE 5: STUDENT TARGET SHEET (3 TARGETS PER STUDENT)

STUDENT NAME: _____________________________ CLASS: __________ DATE: ____________

CURRENT MOCK RESULTS:

Paper 1 (Mock 4): ___/75 (_____%) Grade: _____

Paper 2 (Mock 5): ___/75 (_____%) Grade: _____

Total: ___/150 (_____%) Overall Grade: _____

Target Grade: _____

TARGET 1: TOPIC IMPROVEMENT (知识提升目标)

I will improve my understanding of:

My current score in this topic: ___/___

My target score for next assessment: ___/___

I will achieve this by:

[ ] Revising my notes on this topic

[ ] Completing 5 past paper questions

[ ] Asking my teacher for extra help

[ ] Using online resources

[ ] Peer study with: ________________________

[ ] Other: _________________________________

How I will know I have improved:

TARGET 2: SKILL DEVELOPMENT (技能发展目标)

I will improve my ability to:

Examples of this skill:

[ ] Writing pseudocode under timed conditions

[ ] Completing trace tables accurately

[ ] Writing SQL queries

[ ] Drawing logic circuits

[ ] Answering "Explain" and "Evaluate" questions

[ ] Managing exam time

[ ] Other: _________________________________

I will practise this skill by:

My target: ____

TARGET 3: EXAM TECHNIQUE (考试技巧目标)

I will improve my exam technique by:

Specific actions:

[ ] Reading each question TWICE before answering

[ ] Checking the command word (State/Describe/Explain/Evaluate)

[ ] Showing ALL working for calculation questions

[ ] Planning my answer before writing (especially 15-mark questions)

[ ] Leaving 5 minutes at the end to review

[ ] Not spending more than 1.4 minutes per mark

[ ] Using comments and meaningful names in pseudocode

[ ] Other: _________________________________

How I will practise this:

STUDENT SIGNATURE: __________________________ DATE: ___________

TEACHER SIGNATURE: ___________________________ DATE: ____________

REVIEW DATE: ____________ ACHIEVED? [ ] YES [ ] PARTLY [ ] NOT YET

TEMPLATE 6: REVISION PLAN TEMPLATE

STUDENT NAME: ___________________________ EXAM DATE: ____________

WEEKLY REVISION SCHEDULE:

Day	Topic/Activity	Resources	Time	Done?
Monday				[ ]
Tuesday				[ ]
Wednesday				[ ]
Thursday				[ ]
Friday				[ ]
Saturday				[ ]
Sunday				[ ]

REVISION TECHNIQUES TO USE (复习方法):

[ ] Active recall (write everything you remember, then check)

[ ] Past paper questions (timed)

[ ] Flashcards (key terms and definitions)

[ ] Teach someone else (explain topics to a friend/family member)

[ ] Mind maps (visual connections between topics)

[ ] Video tutorials (watch and take notes)

[ ] Group study (discuss difficult topics)

[ ] Self-testing (use mark schemes to check answers)

[ ] Error log review (learn from past mistakes)

[ ] Formula practice (calculate file sizes, binary conversions)

PRIORITY TOPICS (weakest first):

18. ____________________________________ Time needed: _____ hours
19. ____________________________________ Time needed: _____ hours
20. ____________________________________ Time needed: _____ hours

RESOURCES I WILL USE:

[ ] Past papers (0478/11, 0478/12 for Paper 1; 0478/21, 0478/22 for Paper 2)

[ ] Syllabus revision booklet

[ ] Bilingual glossary

[ ] Pseudocode reference card

[ ] SQL syntax card

[ ] Online practice (specify: _________________________)

[ ] Teacher extra help sessions

[ ] Revision videos (specify: _________________________)

TEMPLATE 7: CONFIDENCE CHECKLIST (RATE 1-5 FOR EACH TOPIC)

STUDENT NAME: ____________________________ DATE: ____________

Instructions: Rate your confidence for each topic from 1 to 5:

1 = I don't understand this at all (完全不懂)

2 = I understand a little but need lots of help (需要很多帮助)

3 = I understand most of this but make some mistakes (大部分懂但会犯错)

4 = I am confident in this topic (比较自信)

5 = I am very confident and can teach others (非常自信能教别人)

PAPER 1 TOPICS:

#	Topic	Rating (1-5)	Evidence/Notes
1	Binary to denary conversion		
2	Denary to binary conversion		
3	Hexadecimal conversion		
4	Two's complement		
5	File size calculations		
6	Sound sampling		
7	Image resolution and colour depth		
8	Lossy vs lossless compression		
9	Error detection (parity, checksum)		
10	Symmetric vs asymmetric encryption		
11	CPU components (ALU, CU, registers)		
12	Fetch-decode-execute cycle		
13	Memory (RAM, ROM, cache)		
14	Secondary storage types		
15	Operating system functions		
16	Compiler vs interpreter		
17	LAN vs WAN		
18	Network topologies		

19	IP addressing		
20	Packet switching		
21	DNS		
22	Security threats and protections		
23	SSL/TLS and HTTPS		
24	Cookies		
25	Sensors and actuators		
26	AI and expert systems		

PAPER 2 TOPICS:

#	Topic	Rating (1-5)	Evidence/Notes
27	Writing pseudocode algorithms		
28	IF...THEN...ELSE structures		
29	FOR loops		
30	WHILE loops		
31	REPEAT...UNTIL loops		
32	Arrays (1D and 2D)		
33	File handling (OPENFILE, etc.)		
34	Functions		
35	Procedures		
36	Parameters and return values		
37	Trace tables		
38	Code reading/debugging		
39	String handling		
40	Database concepts		
41	SQL SELECT queries		
42	SQL WHERE, ORDER BY		
43	SQL aggregate functions		
44	Logic gates (AND, OR, NOT)		
45	Truth tables		
46	Boolean simplification		
47	Logic circuit drawing		

48	15-mark scenario questions		

MY PRIORITY TOPICS (rated 1 or 2):

PART 5: STUDENT REPORT TEMPLATE

END-OF-COURSE STUDENT REPORT

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE Computer Science (0478) -- Year 10 Semester 2

Student Report

STUDENT NAME: ____

CLASS: _________________ CLASS TEACHER: ___________________

DATE: ___________________ ACADEMIC YEAR: 2025-2026

SECTION 1: MOCK EXAMINATION RESULTS

PAPER 1 (Theory) -- 75 marks

Paper	Date	Raw Mark	/75	Percentage	Grade	Change
Mock 1	Y10S1 W17		75			--
Mock 2	Y10S2 W9		75			
Mock 4	Y10S2 W15		75			

Improvement Mock 1 to Mock 4: ______ marks (______% change)

Trend: [ ] Improving [ ] Stable [ ] Declining

PAPER 2 (Problem Solving) -- 75 marks

Paper	Date	Raw Mark	/75	Percentage	Grade	Change
Mock 1	Y10S1 W18		75			--
Mock 3	Y10S2 W10		75			
Mock 5	Y10S2 W16		75			

Improvement Mock 1 to Mock 5: ______ marks (______% change)

Trend: [ ] Improving [ ] Stable [ ] Declining

OVERALL PERFORMANCE (Paper 1 + Paper 2 = 150 marks)

Paper	P1 Mark	P2 Mark	Total	/150	%	Grade
Mock 1				150		
Mock 2+3				150		
Mock 4+5				150		

TARGET GRADE: ___________ PREDICTED GRADE: ___________

SECTION 2: TOPIC MASTERY TRACKER

For each topic, indicate mastery level:

++ = Strong (consistently scores 70%+)

\+ = Good (scores 50-69%)

- = Developing (scores 30-49%)

-- = Needs attention (scores below 30%)

#	Topic	Mock 1	Mock 2/3	Mock 4/5
1	Binary & Hexadecimal			
2	Data Storage Calculations			
3	Sound and Image Representation			
4	Data Transmission			
5	Encryption & Security			
6	CPU Architecture & FDE			
7	Memory & Storage			
8	Software (OS, Utilities, Compilers)			
9	Networks & Topologies			
10	Internet Technologies			
11	Security Threats & Protections			
12	Automated Systems & Sensors			
13	AI and Expert Systems			
14	Emerging Technologies			
15	Algorithm Design			
16	Pseudocode (Selection, Loops)			
17	Arrays & File Handling			
18	Functions & Procedures			
19	Trace Tables & Code Reading			
20	Databases & SQL			
21	Boolean Logic & Circuits			

22	Scenario Questions			

SECTION 3: PROGRESS SUMMARY

Since the beginning of Year 9, this student has:

KNOWLEDGE DEVELOPMENT:

[ ] Demonstrated excellent recall of all syllabus topics

[ ] Shown good understanding of most topics

[ ] Is developing understanding of several topics

[ ] Needs significant support with core concepts

SKILL DEVELOPMENT:

[ ] Writes accurate pseudocode independently

[ ] Completes trace tables with few errors

[ ] Can write SQL queries correctly

[ ] Understands Boolean logic and can complete truth tables

[ ] Can draw logic circuits from Boolean expressions

[ ] Is developing exam technique and time management

EXAM TECHNIQUE:

[ ] Manages time well in examinations

[ ] Reads questions carefully and answers appropriately

[ ] Shows working in calculation questions

[ ] Structures extended answers effectively

[ ] Is improving in exam technique with practice

SECTION 4: STRENGTHS

21. ____

22. ____

23. ____

SECTION 5: DEVELOPMENT AREAS

24. ____

Suggested action: ____

25. ____

Suggested action: ____

26. ____

Suggested action: ____

SECTION 6: EXAM READINESS RATING

READINESS FOR IGCSE EXAMINATION:

[ ] FULLY READY -- Consistently achieving target grade in mocks.

Ready to sit the examination with confidence.

[ ] NEARLY READY -- Achieving close to target grade. A few topics still need consolidation. Continue targeted revision.

[ ] DEVELOPING -- Making progress but not yet at target grade. Intensive revision needed on identified weak areas.

[ ] SIGNIFICANT SUPPORT NEEDED -- Well below target grade. Requires intensive intervention and support plan.

TEACHER COMMENTS:

RECOMMENDED ACTIONS BEFORE EXAMS:

PARENT/CARER: Please sign to acknowledge receipt of this report.

PARENT SIGNATURE: ____________________________ DATE: ____________

TEACHER SIGNATURE: ____________________________ DATE: ____________

HEAD OF DEPARTMENT: ____________________________ DATE: ____________

PART 6: EXAM DAY PREPARATION GUIDE

WHAT TO BRING: EXAM DAY CHECKLIST

ESSENTIAL ITEMS (必须携带):

[ ] BLACK pen (at least 2) -- for writing answers

[ ] BLUE pen (optional) -- some students use for diagrams

[ ] PENCIL (HB) -- for diagrams and rough work

[ ] ERASER -- clean, good quality

[ ] RULER -- for drawing straight lines in diagrams

[ ] STUDENT ID CARD -- required for identification

[ ] EXAM ENTRY STATEMENT -- provided by exams officer

[ ] WATER BOTTLE -- clear bottle, no label

RECOMMENDED ITEMS (建议携带):

[ ] SPARE pens and pencils

[ ] SHARPENER

[ ] TISSUES

[ ] WATCH (not smart watch) -- for time management

[ ] SMALL SNACK -- for energy between papers (eat outside exam room)

FORBIDDEN ITEMS (严禁携带):

[ ] Mobile phone (must be switched off and left with invigilator)

[ ] Smart watch / smart device

[ ] Calculator (NOT allowed for 0478)

[ ] Notes or revision materials

[ ] Dictionary (unless special arrangement approved)

NIGHT BEFORE EXAM:

[ ] Pack your bag with all essential items

[ ] Set two alarms

[ ] Go to bed early (aim for 8+ hours sleep)

[ ] Do a small amount of light revision if it helps you relax

[ ] DO NOT cram all night -- sleep is more important!

MORNING OF EXAM:

[ ] Eat a good breakfast (brain needs energy)

[ ] Arrive at least 30 minutes before the exam starts

[ ] Go to the toilet before entering the exam room

[ ] Stay calm -- deep breaths!

TIME MANAGEMENT STRATEGY

PAPER 1 (Theory)

Total time: 1 hour 45 minutes = 105 minutes

Total marks: 75

Time per mark: 105 / 75 = 1.4 minutes per mark

RECOMMENDED TIMING:

- Quick scan of all questions: 2 minutes
- Section A (short answer, ~30 marks): 35 minutes
- Section B (structured, ~30 marks): 40 minutes
- Section C (extended answer, ~15 marks): 20 minutes
- Review and check: 8 minutes

TIPS:

- Don't spend more than 2 minutes on a 1-mark question
- If stuck, mark the question and MOVE ON
- Leave space to come back to difficult questions
- Use the last 8 minutes to check and complete skipped questions

PAPER 2 (Problem Solving)

Total time: 1 hour 45 minutes = 105 minutes

Total marks: 75

Time per mark: 1.4 minutes per mark

RECOMMENDED TIMING:

- Quick scan: 2 minutes
- Section A (short answer, ~25 marks): 25 minutes
- Section B (code reading/writing, ~35 marks): 45 minutes
- Section C (scenario question, 15 marks): 25 minutes
- Review and check: 8 minutes

TIPS:

- Trace tables: work carefully but don't spend too long

- Scenario question: PLAN for 5 minutes before coding
- Check that all loops and IF statements are closed
- Review your code for syntax errors in the last 8 minutes

"WHAT TO DO IF STUCK" FLOWCHART

```
                ARE YOU STUCK?
             YES          NO
       READ QUESTION      CONTINUE
       ONE MORE TIME      ANSWERING
       +-------+-------+      |
     STILL      UNDER-     |
     STUCK?      STOOD?     |
   +----+----+   +----+----+  |
  YES     NO  YES     NO  |
 WRITE DOWN    | ANSWER    |  |
EVERYTHING   MOVE  IT     RE-  |
 YOU KNOW    TO NEXT       READ |
 ABOUT THE  QUESTION       IT   |
  TOPIC     (COME BACK     SLOWLY|
    |      LATER)         |  |
   +----+----+            |  |
  TRY TO CONNECT          |  |
  YOUR KNOWLEDGE          |  |
  TO THE QUESTION         |  |
   +----+----+            |  |
  MANAGE   STILL          |  |
   TO      CAN'T          |  |
  ANSWER?                 |  |
+----+----+               |  |
YES      NO               |  |
WRITE    GUESS   <---------------+  |
IT     INTELLI-              |
     GENTLY!                 |
+----+----+                  |
 CONTINUE                    |
TO NEXT                      |
QUESTION                     |
```

LAST-MINUTE REVISION PRIORITIES (考试前最后复习重点)

HIGHEST MARK / MOST COMMON TOPICS FOR PAPER 1:

[PRIORITY 1] Fetch-Decode-Execute Cycle (guaranteed question, 4-6 marks)

>> Know ALL registers and their purposes

>> Know the three buses and what they carry

>> Be able to describe the cycle step-by-step

[PRIORITY 2] Network topologies and protocols (4-6 marks)

>> Draw and label star and bus topologies

>> Know the differences between LAN and WAN

>> Understand DNS, HTTP, HTTPS, FTP

[PRIORITY 3] Security (4-6 marks)

>> Know at least 5 security threats

>> Know at least 5 protection methods

>> Be able to match threats to protections

[PRIORITY 4] Data representation calculations (3-4 marks)

>> Binary to denary, denary to binary, binary to hex

>> File size calculations (images and sound)

>> Show your working!

[PRIORITY 5] Emerging technology evaluation (4-6 marks)

>> AI, expert systems, robotics

>> Always give balanced view (advantages AND disadvantages)

HIGHEST MARK / MOST COMMON TOPICS FOR PAPER 2:

[PRIORITY 1] Trace tables (4-6 marks)

>> Set up columns for ALL variables

>> Work through ONE iteration at a time

>> Be careful with loop conditions

[PRIORITY 2] Pseudocode writing (4-8 marks)

>> Practice functions, procedures, loops, arrays

>> Use meaningful variable names

>> Add comments

>> Make sure all structures are closed

[PRIORITY 3] SQL queries (3-6 marks)

>> SELECT, FROM, WHERE, ORDER BY

>> Aggregate functions (COUNT, SUM, AVG)

>> JOIN syntax

[PRIORITY 4] Boolean logic (3-5 marks)

>> Truth tables for AND, OR, NOT, XOR

>> Drawing logic circuits

>> Boolean algebra simplification

[PRIORITY 5] 15-mark scenario question

>> READ the scenario THREE times

>> PLAN before coding (5 minutes)

>> Use modular design (functions/procedures)

>> Include input validation

>> Add comments

>> TEST your solution mentally

PART 7: BILINGUAL GLOSSARY POSTER (QUICK REFERENCE)

This glossary is designed to be printed on A3 or A4 and displayed in the classroom, or provided as a laminated reference card for each student.

EXAM VOCABULARY (考试词汇)

English	Chinese (Simplified)	Definition
Multiple choice	选择题	Question with several possible answers
Structured question	结构化问题	Question with several connected parts
Extended answer	拓展回答	Longer written response (4-6 marks)
Mark scheme	评分标准	Guide used to award marks
Grade boundary	等级分数线	Minimum mark for each grade
Raw mark	原始分数	Actual marks scored
Trace table	跟踪表	Table showing variable value changes
Pseudocode	伪代码	Algorithm in structured English
Flowchart	流程图	Diagram using symbols for algorithm
Scenario question	情景题	Long question based on real-world case
Command word	指令词	Tells you what to do (State/Describe/Explain)
Timed practice	限时练习	Practice under exam time limits
Past paper	历年真题	Exam paper from previous year
Exam technique	考试技巧	Strategies for answering well
Time management	时间管理	Using exam time wisely
Gap analysis	差距分析	Finding weak topics
Mock exam	模拟考试	Practice exam under exam conditions
Revision	复习	Going over work for exam
Consolidation	巩固	Strengthening existing knowledge

DATA REPRESENTATION (数据表示)

Binary	二进制	Number system using 0 and 1 (base-2)
Denary / Decimal	十进制	Number system using 0-9 (base-10)
Hexadecimal	十六进制	Number system 0-9, A-F (base-16)
Two's complement	补码	Method for negative binary numbers
Bit	比特	Smallest unit of data (0 or 1)
Byte	字节	8 bits
Kilobyte (KB)	千字节	1024 bytes
Megabyte (MB)	兆字节	1024 KB
Gigabyte (GB)	吉字节	1024 MB
Terabyte (TB)	太字节	1024 GB
Pixel	像素	Single point in a digital image
Colour depth	颜色深度	Bits per pixel
Resolution	分辨率	Width x height in pixels
Sample rate	采样率	Samples per second (Hz)
Sample resolution	采样分辨率	Bits per sound sample
Metadata	元数据	Data about data
Lossy compression	有损压缩	Removes data permanently
Lossless compression	无损压缩	Reduces size without data loss
Parity check	奇偶校验	Error detection using extra bit
Checksum	校验和	Error detection using calculated value
Encryption	加密	Converting data to secure code
Plain text	明文	Unencrypted readable data
Cipher text	密文	Encrypted data
Symmetric encryption	对称加密	Same key for encrypt and decrypt
Asymmetric encryption	非对称加密	Different keys (public/private)

HARDWARE AND SOFTWARE (硬件与软件)

CPU	中央处理器	"Brain" of the computer
ALU	算术逻辑单元	Performs calculations and logic
CU	控制单元	Manages instruction execution
Register	寄存器	Small fast memory in CPU
Program Counter (PC)	程序计数器	Holds address of next instruction
Memory Address Reg (MAR)	内存地址寄存器	Holds memory address being accessed
Memory Data Reg (MDR)	内存数据寄存器	Holds data being transferred
Accumulator (ACC)	累加器	Stores calculation results

Current Instr Reg (CIR)	当前指令寄存器	Holds instruction being decoded
Fetch-decode-execute	取指-译码-执行	Cycle CPU follows to run programs
RAM	随机存取存储器	Volatile read/write memory
ROM	只读存储器	Non-volatile read-only memory
Cache	高速缓存	Fast memory between CPU and RAM
Virtual memory	虚拟内存	Using disk space as extra RAM
HDD	硬盘驱动器	Magnetic storage with moving parts
SSD	固态硬盘	Flash memory, no moving parts
Operating system	操作系统	Manages computer resources
Compiler	编译器	Translates whole program at once
Interpreter	解释器	Translates line by line
Assembler	汇编器	Converts assembly to machine code
Freeware	免费软件	Free software
Shareware	共享软件	Try before you buy
Open source	开源	Source code available to modify

INTERNET AND SECURITY (互联网与安全)

LAN	局域网	Local Area Network
WAN	广域网	Wide Area Network
Star topology	星型拓扑	All devices connect to central point
Bus topology	总线拓扑	All devices connect to single cable
Router	路由器	Connects networks, routes packets
Switch	交换机	Connects devices in a LAN
IP address	IP 地址	Unique network identifier
MAC address	MAC 地址	Unique hardware identifier
Packet switching	分组交换	Data split into packets for sending
Protocol	协议	Rules for data communication
DNS	域名系统	Translates URLs to IP addresses
HTTP	超文本传输协议	Web page transfer protocol
HTTPS	安全超文本传输协议	Encrypted HTTP
SSL/TLS	安全传输层协议	Encryption for secure transmission
HTML	超文本标记语言	Language for web pages
Phishing	钓鱼攻击	Fake emails to steal information
Pharming	网址嫁接	Redirecting to fake websites
Hacking	黑客攻击	Unauthorised system access
Firewall	防火墙	Blocks unauthorised access

Antivirus	杀毒软件	Detects and removes malware
Authentication	身份验证	Confirming identity
Cookie	Cookie	Small file stored by browser
Malware	恶意软件	Malicious software

AUTOMATED TECHNOLOGIES (自动化技术)

Sensor	传感器	Detects physical conditions
Actuator	执行器	Produces physical movement
ADC	模数转换器	Analogue to Digital Converter
DAC	数模转换器	Digital to Analogue Converter
Microprocessor	微处理器	Single-chip CPU
Robotics	机器人技术	Design and use of robots
Artificial intelligence	人工智能	Systems mimicking human intelligence
Machine learning	机器学习	Systems that learn from data
Expert system	专家系统	Mimics human expertise
Knowledge base	知识库	Facts and information
Inference engine	推理引擎	Applies rules to knowledge
Rules base	规则库	IF-THEN rules
3D printing	3D 打印	Creating physical objects from digital
Virtual reality	虚拟现实	Immersive computer-generated world
Augmented reality	增强现实	Digital overlay on real world

PROGRAMMING (编程)

Algorithm	算法	Step-by-step problem solution
Decomposition	分解	Breaking problem into smaller parts
Abstraction	抽象	Removing unnecessary detail
Variable	变量	Named storage for data
Constant	常量	Value that does not change
Array	数组	Collection of data items
1D array	一维数组	Single list of values
2D array	二维数组	Table of values (rows/columns)
Selection	选择	Decision making (IF, CASE)
Iteration	迭代	Repeating instructions
FOR loop	FOR 循环	Count-controlled loop
WHILE loop	WHILE 循环	Condition-controlled (check first)
REPEAT UNTIL	直到循环	Condition-controlled (check last)
Function	函数	Returns a value

Procedure	过程	Performs a task
Parameter	参数	Value passed to subprogram
Return value	返回值	Value sent back from function
Local variable	局部变量	Accessible only in its subprogram
Global variable	全局变量	Accessible from anywhere
Library routine	库例程	Pre-written common code
String handling	字符串处理	Operations on text
File handling	文件处理	Reading/writing files
CSV	逗号分隔值	Comma Separated Values format
EOF	文件结尾	End Of File
Trace table	跟踪表	Shows variable changes step by step

DATABASES AND BOOLEAN LOGIC (数据库与布尔逻辑)

Database	数据库	Organised data collection
Flat-file database	平面文件数据库	Single table database
Relational database	关系型数据库	Multiple linked tables
Table	表	Collection of records
Record	记录	One row in a table
Field	字段	One column in a table
Primary key	主键	Unique record identifier
Foreign key	外键	Links to primary key in another table
ER diagram	实体关系图	Visual database structure
Normalisation	规范化	Organising to reduce redundancy
SQL	结构化查询语言	Database language
SELECT	查询语句	Retrieve data
WHERE	条件子句	Filter records
ORDER BY	排序子句	Sort results
Wildcard	通配符	Pattern matching character
Aggregate function	聚合函数	COUNT, SUM, AVG, MAX, MIN
JOIN	连接	Combine tables
Boolean	布尔	True/False logic
Logic gate	逻辑门	Electronic Boolean circuit
AND gate	与门	True only if both inputs true
OR gate	或门	True if either input true
NOT gate	非门	Inverts input
NAND gate	与非门	NOT-AND combination

NOR gate	或非门	NOT-OR combination
XOR gate	异或门	True if inputs differ
Truth table	真值表	All input/output combinations
Boolean expression	布尔表达式	Mathematical Boolean expression
Logic circuit	逻辑电路	Connected logic gates diagram

COMMAND WORDS FOR EXAMS (考试指令词)

State / Name / Give	陈述/命名/给出	Brief answer, no explanation needed
Describe	描述	Give details of how it works
Explain	解释	Give reasons why
Compare	比较	Give similarities AND differences
Evaluate	评估	Give advantages AND disadvantages
Discuss	讨论	Consider different viewpoints
Draw	画出	Produce a diagram
Write	写出	Produce pseudocode/SQL
Complete	完成	Fill in missing information
Calculate	计算	Show your working

EMERGENCY REMINDERS (考试紧急提醒)

>> Don't panic! Breathe deeply. (不要慌张!深呼吸。)

>> Read the question twice. (读两遍题目。)

>> Show all working. (展示所有过程。)

>> Never leave a blank. (不要留空白。)

>> 1.4 minutes per mark. (每分 1.4 分钟。)

>> Check your answers at the end. (最后检查答案。)

>> You know more than you think. (你掌握的知识比你以为的多。)

>> Trust your revision. (相信你的复习。)

END OF ASSESSMENT PACK

Document Information

File: 03_VPA_CS_Y10S2_Assessment_Pack.txt

Course: CIE IGCSE Computer Science (0478)

School: Victoria Park Academy, Shenzhen

Contents:

- Mock Exam 2 Paper 1 (Week 9) + Mark Scheme + Grade Boundaries
- Mock Exam 3 Paper 2 (Week 10) + Mark Scheme + Grade Boundaries
- Mock Exam 4 Paper 1 (Week 15) + Mark Scheme + Grade Boundaries
- Mock Exam 5 Paper 2 (Week 16) + Mark Scheme + Grade Boundaries
- Gap Analysis Templates (4 templates)

- Target Setting Sheets (3 templates)
- Student Report Template
- Exam Day Preparation Guide
- Bilingual Glossary Poster (50+ terms)

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